
MATH 563: Image De-Blurring and De-Noising

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1 Introduction and Problem Formulation

Let $x \in \mathbb{R}^n$ denote a (vectorized) image with $n = N^2$ pixels, and $b \in \mathbb{R}^n$ the observable image from which we wish to reconstruct x . We assume b is an affine transformation of x through a linear blur operator K and additive noise w according to $b = Kx + w$.

1.1 Objective functions

The problem of image de-blurring in the presence of noise can be expressed as an optimization problem whose objective combines a measure of data fidelity (reconstruction error) with a penalty designed to encourage local smoothness while preserving sharp edge boundaries.

Model 1: ℓ_1 reconstruction. Target exact recovery by enforcing residual sparsity $Kx - b$ via ℓ_1 loss:

$$\min_{x \in \mathbb{R}^n} \{ \|Kx - b\|_1 + \gamma \|Dx\|_{\text{iso}} + \delta_C(x) \}. \quad (1)$$

Here, $D : \mathbb{R}^n \rightarrow \mathbb{R}^{2n}$ is the discrete gradient operator and $\|\cdot\|_{\text{iso}}$ the total variation penalty:

$$\|Dx\|_{\text{iso}} = \sum_{i=1}^n \sqrt{(D_1 x_i)^2 + (D_2 x_i)^2},$$

where $\delta_C(x)$ is the indicator function over the unit box $C := \{x \in \mathbb{R}^n \mid 0 \leq x_i \leq 1\}$, restricting each pixel to the physical range $[0, 1]$.

Model 2: ℓ_2 reconstruction. In contrast, consider the standard ℓ_2 distance:

$$\min_{x \in \mathbb{R}^n} \{ \|Kx - b\|_2^2 + \gamma \|Dx\|_{\text{iso}} + \delta_C(x) \}. \quad (2)$$

2 Derivation of Proximal Operators

2.1 Box projection operator (boxProx)

Let $f(x) = \delta_C(x)$ denote the indicator function on the convex set $C = \{x \in \mathbb{R}^n \mid 0 \leq x_i \leq 1\}$ (the unit box in $\mathbb{R}^n$). The proximal operator reduces to the Euclidean projection onto C :

$$\text{prox}_{\lambda \delta_C}(x) = \arg \min_{u \in C} \frac{1}{2\lambda} \|u - x\|_2^2 = \text{proj}_C(x). \quad (3)$$

Since the constraint is separable with respect to the components $x_1, \dots, x_n$, the projection is applied component-wise such that the i -th component of $\text{proj}_C(x)$ is simply:

$$(\text{proj}_C(x))_i := \max\{0, \min\{1, x_i\}\}. \quad (4)$$

2.2 ℓ_1 -Norm proximal operator (11Prox)

Let $f(x) = \|x\|_1$ such that the proximal operator is defined as:

$$\text{prox}_{\lambda\|\cdot\|_1}(x) := \arg \min_u \left\{ \|u\|_1 + \frac{1}{2\lambda} \|u - x\|_2^2 \right\}.$$

This is separable. Take the subgradient with respect to the i -th component and set the result to zero:

$$0 \in \partial|x_i| + \frac{1}{\lambda}(u_i - x_i) \implies u_i = x_i - \lambda\partial|x_i|.$$

The subgradient $\partial|x_i|$ is well-known, and hence the solution for the i -th coordinate is given by the soft-thresholding operator:

$$S_\lambda(x_i) := \text{sgn}(x_i) \max\{|x_i| - \lambda, 0\} \implies (\text{prox}_{\lambda\|\cdot\|_1}(x))_i = S_\lambda(x_i).$$

2.3 Isotropic TV proximal operator (isoProx)

Let $f(u) = \gamma \sum_i \sqrt{(u_{1,i})^2 + (u_{2,i})^2}$, where $u = (u_1, u_2)$ represents the horizontal and vertical gradients. The problem is separable by pixel index i . For a given pixel, partition u and x by $u_i = ((u_1)_i, (u_2)_i)$ and $x_i = ((x_1)_i, (x_2)_i)$. Then the i -th component of the proximal operator is:

$$(\text{prox}_{\lambda f}(x))_i := \arg \min_{u_i} \left\{ \gamma \|u_i\|_2 + \frac{1}{2\lambda} \|u_i - x_i\|_2^2 \right\}.$$

Take the subgradient with respect to the vector u_i :

$$0 \in \gamma \partial \|u_i\|_2 + \frac{1}{\lambda}(u_i - x_i).$$

If $u_i \neq 0$ then $\partial \|u_i\|_2 = \frac{u_i}{\|u_i\|_2}$. Substituting this in, the vectors u_i and x_i must be collinear, i.e. $u_i = cx_i$ for some $c > 0$. Solving for u_i yields the block soft-thresholding operator:

$$(\text{prox}_{\lambda f}(x))_i = \max \left\{ 1 - \frac{\lambda\gamma}{\|x_i\|_2}, 0 \right\} \cdot x_i.$$

3 Algorithmic Frameworks

To solve (1) and (2) we cast them into the form $\min_x f(x) + g(Ax)$. Define $A := [K^\top \ D^\top]^\top \in \mathbb{R}^{3n \times n}$, where $K \in \mathbb{R}^{n \times n}$ is the chosen blur kernel and $D \in \mathbb{R}^{2n \times n}$ is the discrete gradient operator. Let $f : \mathbb{R}^n \rightarrow \bar{\mathbb{R}}$ and $g : \mathbb{R}^{3n} \rightarrow \mathbb{R}$ be defined as

$$f(x) = \delta_C(x), \quad g(z) = g(z_1, z_2) = g_1(z_1) + \gamma \|z_2\|_{\text{iso}}, \quad (5)$$

where $z := [z_1^\top, z_2^\top]^\top$. Here, $z_1 \in \mathbb{R}^n$ corresponds to the residuals $Kx - b$ and $z_2 \in \mathbb{R}^{2n}$ corresponds to the penalty argument Dx in models (1) and (2). Depending on the chosen model, $g_1(z_1)$ is either the ℓ_1 or (squared) ℓ_2 norm of $z_1 - b$.

3.1 Proximal mappings for the split functions

- **Model 1 (ℓ_1 loss).** We have $g_1(z_1) = \|z_1 - b\|_1$. By the translation property of proximal operators (i.e. $\text{prox}_{h(\cdot - c)}(x) = c + \text{prox}_h(x - c)$), it follows that the proximal operator of $\|z_1 - b\|_1$ is equivalent to the proximal operator of the ℓ_1 norm derived in Section 2.2, shifted by the observation vector b . To wit:

$$\text{prox}_{\lambda g_1}(z_1) = b + S_\lambda(z_1 - b),$$

where $S_\lambda(\cdot)$ is the 1D soft-thresholding operator defined in Section 2.2.

- **Model 2 (ℓ_2 loss).** We have $g_1(z_1) = \|z_1 - b\|_2^2$. The proximal operator is obtained by setting the the gradient of the proximal objective to zero:

$$\text{prox}_{\lambda g_1}(z_1) = \arg \min_{u_1} \left\{ \|u_1 - b\|_2^2 + \frac{1}{2\lambda} \|u_1 - z_1\|_2^2 \right\},$$

hence

$$0 = 2(u_1 - b) + \frac{1}{\lambda}(u_1 - z_1) \implies \text{prox}_{\lambda g_1}(z_1) = \frac{z_1 + 2\lambda b}{1 + 2\lambda}.$$

Finally, the proximal mapping for the regularization term $g_2(z_2) = \gamma \|z_2\|_{\text{iso}}$ is precisely that given by the isotropic TV proximal operator `isoProx` derived in Section 2.3.

3.2 Algorithm 1: Primal Douglas-Rachford Splitting

The primal DR algorithm minimizes $f(x) + g(y)$ subject to the constraint $Ax = y$. We absorb the constraint $Ax = y$ into the objective by introducing an indicator function:

$$\min_{x,y} f(x) + g(y) + \delta_0 \left([A \quad -I] \begin{bmatrix} x \\ y \end{bmatrix} \right), \quad (6)$$

where $\delta_0(v) = 0$ if $v = 0$ and $+\infty$ otherwise. This formulation is equivalent to the original constrained problem because δ_0 enforces $Ax - y = 0$ exactly. The first-order optimality condition for (6) is

$$0 \in \underbrace{\partial f(x) + \partial g(y)}_{\mathcal{A}(x,y)} + \underbrace{\partial \delta_0 \left([A \quad -I] \begin{bmatrix} x \\ y \end{bmatrix} \right)}_{\mathcal{B}(x,y)}. \quad (7)$$

Since $\mathcal{A}(x, y) = \partial f(x) + \partial g(y)$ is separable in x and y , its resolvent decouples into independent proximal operator evaluations:

$$(I + t\mathcal{A})^{-1}(\bar{x}, \bar{y}) = (\text{prox}_{tf}(\bar{x}), \text{prox}_{tg}(\bar{y})).$$

The operator $\mathcal{B}$ encodes the constraint $Ax = y$ through the indicator term. Since the subdifferential of an indicator function is the normal cone to the underlying constraint set, the resolvent $(I + t\mathcal{B})^{-1}$ is the Euclidean projection of $(\bar{x}, \bar{y})$ onto the subspace $\{(x, y) : Ax = y\}$. Therefore, $(I + t\mathcal{B})^{-1}(\bar{x}, \bar{y})$ is obtained by solving

$$\min_{x,y} \frac{1}{2} \|x - \bar{x}\|^2 + \frac{1}{2} \|y - \bar{y}\|^2 \quad \text{subject to } Ax = y.$$

Substituting $y = Ax$ reduces this to

$$\min_x \frac{1}{2} \|x - \bar{x}\|^2 + \frac{1}{2} \|Ax - \bar{y}\|^2.$$

Its first-order optimality condition is

$$0 = (x - \bar{x}) + A^T(Ax - \bar{y}),$$

so

$$(I + A^T A)x = \bar{x} + A^T \bar{y}.$$

Hence

$$x = (I + A^T A)^{-1}(\bar{x} + A^T \bar{y}), \quad y = Ax. \quad (8)$$

3.3 Algorithm 2: Primal-Dual Douglas-Rachford Splitting

The Primal-Dual Douglas-Rachford algorithm extends the primal splitting of Section 3 by operating jointly on the primal variable $x \in \mathbb{R}^n$ and dual variable $z \in \mathbb{R}^{3n}$. The joint first-order optimality conditions for both the primal problem and its dual reduce to the inclusion

$$0 \in \underbrace{\begin{pmatrix} \partial f(x) \\ \partial g^*(z) \end{pmatrix}}_{\mathcal{A}(x,z)} + \underbrace{\begin{pmatrix} 0 & A^\top \\ -A & 0 \end{pmatrix} \begin{pmatrix} x \\ z \end{pmatrix}}_{\mathcal{B}(x,z)}. \quad (9)$$

Since $\mathcal{A}(x, z)$ is separable, its resolvent decouples as $(I + t\mathcal{A})^{-1}(\bar{x}, \bar{z}) = (\text{prox}_{tf}(\bar{x}), \text{prox}_{tg^*}(\bar{z}))$, where $\text{prox}_{tf}(\bar{x}) = \text{proj}_C(\bar{x})$ is the box projection from Section 2.1, and $\text{prox}_{tg^*}(\bar{z})$ is computed via Moreau's decomposition $\text{prox}_{tg^*}(\bar{z}) = \bar{z} - t \cdot \text{prox}_{g/t}(\bar{z}/t)$ using the block-wise proximal operators of Section 3.1. The operator $\mathcal{B}$ is skew-symmetric, so its resolvent $(I + t\mathcal{B})^{-1}(\bar{p}, \bar{q})$ is obtained by solving

$$\begin{pmatrix} I & tA^\top \\ -tA & I \end{pmatrix} \begin{pmatrix} w \\ v \end{pmatrix} = \begin{pmatrix} \bar{p} \\ \bar{q} \end{pmatrix},$$

which gives

$$w = (I + t^2 A^\top A)^{-1}(\bar{p} + tA^\top \bar{q}), \quad v = \bar{q} - tAw, \quad (10)$$

and is again solved efficiently in the Fourier domain as in Algorithm 2 (Appendix E).

3.4 Algorithm 3: Alternating Direction Method of Multipliers

ADMM is Douglas–Rachford splitting applied to the dual problem. In the present setting it enforces the constraints $x = u$ and $Ax = y$ through dual variables w and z , with primal residuals

$$r_1^k = x^k - u^k, \quad r_2^k = Ax^k - y^k.$$

ℓ_1 and ℓ_2^2 models. We begin from

$$\min_x \phi(Kx - b) + \delta_S(x) + \gamma \|Dx\|_{\text{iso}}, \quad S = [0, 1]^n,$$

where

$$\phi(\xi) = \begin{cases} \|\xi\|_1, & \ell_1 \text{ model}, \\ \frac{1}{2} \|\xi\|_2^2, & \ell_2^2 \text{ model}. \end{cases}$$

Introduce the split variables $u = x$, $y_1 = Kx$, $(y_2, y_3) = Dx$, and write $y = (y_1, y_2, y_3)$. Then

$$\min_{x, u, y} \delta_S(u) + \phi(y_1 - b) + \gamma \|(y_2, y_3)\|_{\text{iso}} \quad \text{s.t.} \quad x = u, \quad Ax = y,$$

with $A = [K^\top \ D^\top]^\top$.

Overview of the significance of variables The auxiliary variable u enforces the box constraint $x \in [0, 1]^n$. This ensures valid pixel intensities. The split variable $y = (y_1, y_2, y_3)$ decouples the data fidelity and regularization terms, where $y_1 = Kx$ corresponds to the blurred image and $(y_2, y_3) = Dx$ represent horizontal and vertical gradients. The dual variables w and z enforce consistency of these splits, driving $x = u$ and $Ax = y$ through the residuals, thereby enabling efficient alternating minimization.

Augmented Lagrangian With dual variables w and z , the augmented Lagrangian is

$$\begin{aligned} \mathcal{L}(x, u, y, w, z) = & \delta_S(u) + \phi(y_1 - b) + \gamma \|(y_2, y_3)\|_{\text{iso}} - w^\top (x - u) - z^\top (Ax - y) \\ & + \frac{t}{2} \|x - u\|_2^2 + \frac{t}{2} \|Ax - y\|_2^2. \end{aligned}$$

Iterations At iteration k , ADMM alternates

$$\begin{aligned} x^k &= \arg \min_x \mathcal{L}(x, u^{k-1}, y^{k-1}, w^{k-1}, z^{k-1}), \\ u^k &= \arg \min_u \mathcal{L}(x^k, u, y^{k-1}, w^{k-1}, z^{k-1}), \\ y^k &= \arg \min_y \mathcal{L}(x^k, u^k, y, w^{k-1}, z^{k-1}), \end{aligned}$$

followed by the updates to the multipliers

$$w^k = w^{k-1} + t(x^k - u^k), \quad z^k = z^{k-1} + t(Ax^k - y^k).$$

An over-relaxed version replaces x^k and Ax^k in the u - and y -updates by $\rho x^k + (1 - \rho)u^{k-1}$ and $\rho Ax^k + (1 - \rho)y^{k-1}$, respectively, with $\rho \in (0, 2)$.

x -update Completing squares gives

$$x^k = \arg \min_x \frac{t}{2} \left\| x - u^{k-1} + \frac{w^{k-1}}{t} \right\|_2^2 + \frac{t}{2} \left\| Ax - y^{k-1} + \frac{z^{k-1}}{t} \right\|_2^2,$$

hence

$$(I + A^\top A)x^k = u^{k-1} - \frac{w^{k-1}}{t} + A^\top \left(y^{k-1} - \frac{z^{k-1}}{t} \right).$$

Since K is convolutional and D is circulant under periodic boundary conditions, this linear system is solved efficiently in the Fourier domain.

u -update. Since $f = \delta_S$, its proximal map is projection onto $S = [0, 1]^n$:

$$u^k = \arg \min_u \delta_S(u) + \frac{t}{2} \left\| u - \left(x^k + \frac{w^{k-1}}{t} \right) \right\|_2^2 = \Pi_{[0, 1]^n} \left(x^k + \frac{w^{k-1}}{t} \right).$$

y-update. Let $v^k = Ax^k + z^{k-1}/t$, with block decomposition $v^k = (v_1^k, v_2^k, v_3^k)$. The minimization separates componentwise.

For the data-fidelity block,

$$y_1^k = \arg \min_{y_1} \phi(y_1 - b) + \frac{t}{2} \|y_1 - v_1^k\|_2^2 \iff y_1^k = \begin{cases} b + \text{soft}(v_1^k - b, \frac{1}{t}), & \ell_1, \\ \frac{t v_1^k + b}{t + 1}, & \ell_2. \end{cases}$$

For the regularization block,

$$(y_2^k, y_3^k) = \arg \min_{y_2, y_3} \gamma \|(y_2, y_3)\|_{\text{iso}} + \frac{t}{2} \|(y_2, y_3) - (v_2^k, v_3^k)\|_2^2,$$

whose solution is the isotropic shrinkage

$$(y_2^k, y_3^k) = \max\left(1 - \frac{\gamma}{t\|(v_2^k, v_3^k)\|}, 0\right) (v_2^k, v_3^k).$$

Dual updates. The multipliers accumulate the primal residuals

$$w^k = w^{k-1} + t(x^k - u^k), \quad z^k = z^{k-1} + t(Ax^k - y^k).$$

3.4.1 Convergence

If the functions f and g are proper, closed, and convex, and a saddle point exists, then ADMM converges under the standard assumptions of Douglas–Rachford splitting. In particular, the primal residuals vanish asymptotically and the iterates converge to a solution of the split problem. For a classical proof see [2] and for the Lyapunov-based argument used in this report, see the appendix.

3.5 Algorithm 4: Chambolle-Pock

The Chambolle-Pock algorithm is a first-order primal-dual method designed to solve saddle-point problems of the form:

$$\min_x \max_y \{ \langle Ax, y \rangle + f(x) - g^*(y) \}, \quad (11)$$

where g^* is the Fenchel conjugate of g . The Chambolle-Pock algorithm avoids the costly inversion of the linear system $(I + A^\top A)^{-1}$, and instead requires matrix-vector multiplications between A and $A^\top$. To map the ℓ_1 model (1) and ℓ_2 model (2) to the Chambolle-Pock formulation (11), we first recall the form of the canonical primal problem:

$$\min_x \{ f(x) + g(Ax) \}. \quad (12)$$

One can express g in terms of its Fenchel conjugate: $g(Ax) = \sup_y \{ \langle Ax, y \rangle - g^*(y) \}$, and substituting this expression for g into the primal problem (12) precisely recovers (11).

Explicit algorithmic specification. The Chambolle-Pock algorithm is outlined “Algorithm 4” of the project description, copied here in Algorithm 4 in Appendix E. As shown in (5), we have $f(x) = \delta_C(x)$ and $A = [K^\top \ D^\top]^\top$, while g is the sum of the data loss term (either the ℓ_1 or ℓ_2 loss) and the total variation penalty. To solve (11) via Chambolle-Pock (Algorithm 4), we express the proximal operators for f and g^* in (11).

Primal update (x-step). The primal update involves the f term in (11), namely $f(x) = \delta_C(x)$. As derived in Section 2.1, the proximal operator of $\delta_C(x)$ is the Euclidean projection onto C , independent of the chosen step size t :

$$x^k \leftarrow \text{prox}_{t f}(x^{k-1} - tA^\top y^k) = \text{proj}_C(x^{k-1} - tA^\top y^k), \quad (13)$$

where $\text{proj}_C(\cdot)$ is the component-wise bounding to the range $[0, 1]$ described in Section 2.1.

Dual update (y -step). The dual update relies on evaluating $\text{prox}_{sg^*}(\cdot)$. Since g is explicitly specified but g^* is not, we apply Moreau’s decomposition:

$$\begin{aligned} v &= \text{prox}_{sg^*}(v) + s \cdot \text{prox}_{(g^*)^*/s}(v/s), = \text{prox}_{sg^*}(v) + s \cdot \text{prox}_{g/s}(v/s), \\ &\iff \text{prox}_{sg^*}(v) = v - s \cdot \text{prox}_{g/s}(v/s). \end{aligned} \quad (14)$$

Let $v^k = y^{k-1} + sAz^{k-1}$ denote the argument passed to $\text{prox}_{sg^*}(\cdot)$ and partition v^k into its data fidelity component $v_1^k \in \mathbb{R}^n$ and its gradient components $v_2^k \in \mathbb{R}^{2n}$. For the data fidelity term $g_1(y_1)$, we evaluate $\text{prox}_{g_1/s}(v_1^k/s)$ under the two models:

- **Model 1 (ℓ_1 loss).** We have $g_1(u) = \|u - b\|_1$. As argued in Section 3.1, this proximal operator is the shifted soft-thresholding operator:

$$\text{prox}_{g_1/s}(v_1^k/s) = b + S_{1/s}(v_1^k/s - b). \quad (15)$$

- **Model 2 (ℓ_2 loss).** We have $g_1(u) = \|u - b\|_2^2$. Take the subgradient and set it to zero:

$$\text{prox}_{g_1/s}(v_1^k/s) = \frac{v_1^k + 2b}{s + 2}. \quad (16)$$

The regularization term $g_2(y_2) = \gamma\|y_2\|_{\text{iso}}$ applies to both models, and from Section 2.3 we know $\text{prox}_{g_2/s}(v_2^k/s) = \text{isoProx}_{\gamma/s}(v_2^k/s)$. Therefore, the full dual update is explicitly computed as:

$$y^k \leftarrow v^k - s \begin{bmatrix} \text{prox}_{g_1/s}(v_1^k/s) \\ \text{prox}_{g_2/s}(v_2^k/s) \end{bmatrix} = v^k - s \begin{bmatrix} \text{prox}_{g_1/s}(v_1^k/s) \\ \text{isoProx}_{\gamma/s}(v_2^k/s) \end{bmatrix},$$

where $\text{prox}_{g_1/s}(\cdot)$ is either (15) under the ℓ_1 model or (16) under the ℓ_2 model. The convergence of the Chambolle-Pock algorithm is presented in Appendix D.1.

4 Quality Metrics

Since the restoration task involves *both* denoising and deblurring, a fair evaluation metric should reward both intensity fidelity and recovery of sharp geometric structure. As such, we use a balanced composite score built from two main components - a *denoising block* in the image domain and a *deblurring block* in the gradient domain.

PSNR - For a ground-truth image $x \in \mathbb{R}^{N \times N}$ and reconstruction $b \in \mathbb{R}^{N \times N}$, the mean-squared error is

$$\text{MSE}(x, b) = \frac{1}{N^2} \|x - b\|_F^2.$$

Since all images are normalized to $[0, 1]$, the peak signal-to-noise ratio is

$$\text{PSNR}(x, b) = 10 \log_{10} \left(\frac{1}{\text{MSE}(x, b)} \right) = 10 \log_{10} \left(\frac{N^2}{\|x - b\|_F^2} \right).$$

A larger PSNR indicates smaller pixelwise error. However, PSNR is fundamentally intensity-based, so it is highly sensitive to noise but comparatively less sensitive to blur, since a blurred image may still remain close in Euclidean distance to the original [5].

SSIM - The structural similarity index measures similarity in luminance, contrast, and local structure. For two local patches $u, v \in \mathbb{R}^n$, let μ_u, μ_v denote their local means, σ_u^2, σ_v^2 their local variances, and σ_{uv} their covariance [6], [10]. Then

$$\text{SSIM}(u, v) = \frac{(2\mu_u\mu_v + C_1)(2\sigma_{uv} + C_2)}{(\mu_u^2 + \mu_v^2 + C_1)(\sigma_u^2 + \sigma_v^2 + C_2)}.$$

The full-image SSIM is obtained by averaging this quantity over all local windows. Unlike PSNR, SSIM is sensitive to local structural distortions, but because it is still computed on image intensities, it tends to reward smooth reconstructions and may not fully penalize loss of edge sharpness. A comparison between the PSNR and SSIM has been provided here [3]

Local Monge–Kantorovich discrepancy - To capture local geometric mismatch, patches using a discrete optimal transport-type cost has been compared. For a patch Ω with pixel locations $\{p_i\}_{i=1}^m$, we associate two nonnegative discrete measures

$$\mu = \sum_{i=1}^m a_i \delta_{p_i}, \quad \nu = \sum_{j=1}^m b_j \delta_{p_j},$$

where (a_i) and (b_j) are *normalized* patch intensities ensuring equal total mass. The quadratic cost may be expressed as

$$W_2^2(\mu, \nu) = \min_{\pi \geq 0} \sum_{i,j=1}^m \|p_i - p_j\|^2 \pi_{ij}, \quad \text{s.t.} \quad \sum_j \pi_{ij} = a_i, \quad \sum_i \pi_{ij} = b_j.$$

In practice, this is evaluated patchwise and averaged over all sliding windows. To retain sensitivity to absolute intensity differences, we augment the transport term by a relative mass mismatch and a local ℓ_1 error, giving

$$D_{\text{patch}} = W_2^2(\mu, \nu) + \lambda_{\text{mass}} \frac{|m_\mu - m_\nu|}{\max(m_\mu, m_\nu, \varepsilon)} + \lambda_1 \|u - v\|_{1, \text{mean}},$$

where m_μ, m_ν are the (unnormalized) patch masses. Averaging over all patches yields the final transport discrepancy D_{OT} . This metric captures local spatial rearrangement while remaining sensitive to brightness mismatch.

Gradient-domain deblurring metrics - Blur primarily destroys high-frequency content and attenuates edges. To explicitly measure sharpness recovery, we also evaluate the reconstruction in the gradient domain. Writing

$$\nabla x = (D_x x, D_y x), \quad g(x) = \|\nabla x\|_2 = \sqrt{(D_x x)^2 + (D_y x)^2},$$

we compare $g(x)$ and $g(\hat{x})$ using two quantities: S

$$\text{PSNR}_\nabla = \text{PSNR}(g(x), g(\hat{x})), \quad D_{\text{OT}, \nabla} = D_{\text{OT}}(g(x), g(\hat{x})).$$

The first measures gradient fidelity in an average squared sense, while the second measures geometric mismatch between local edge patterns.

Balanced composite score - To avoid biasing the evaluation toward either denoising or deblurring alone, we define two block scores

$$Q_{\text{denoise}} = \omega_P S_{\text{PSNR}} + \omega_S S_{\text{SSIM}}, \quad Q_{\text{deblur}} = \omega_{\nabla P} S_{\nabla \text{PSNR}} + \omega_{\nabla T} S_{\nabla \text{OT}}.$$

Each component is normalized to lie in $[0, 1]$, with larger values indicating better quality. The final balanced score is

$$Q = \omega_D Q_{\text{denoise}} + \omega_B Q_{\text{deblur}}, \quad 0 \leq Q \leq 1.$$

In experiments, we take $\omega_D = \omega_B = \frac{1}{2}$, ensuring equal contribution from denoising and deblurring quality. This avoids over-rewarding oversmoothed reconstructions and provides a more faithful overall assessment. This is particularly useful when image degradation induces local geometric distortions such as displaced or smeared edges. It is however important to note that a full OT-distance has not been computed. This has been proposed as a heuristic geometric distance and the word *metric* is meant literally.

5 Software Package

The algorithms developed in this work are packaged as a MATLAB application comprising a unified entry-point function and an optional graphical interface. The entry-point `deblur` (`problem`, `algorithm`, `xinitial`, `kernel`, `b`, `i`) accepts a blurred observation b , a point-spread function, an optional initial guess, and a parameter struct `i`; it returns the reconstructed image x . Four solvers are available (PDR, PDDR, ADMM, CP) and both the ℓ_1 and ℓ_2 fidelity variants are supported. Default parameter values, including the evidence-based choices of γ , are summarised in Table 1. Full API documentation, example code, and installation instructions are provided in Appendix F.

6 Experiments

To evaluate the efficacy of the four splitting methods, we conduct tests across varying noise profiles and blur kernels. Unless otherwise specified in a specific experimental section, the package defaults to the parameters outlined in Table 1. A exploration justifying the default setting of the Chambolle-Pock step-sizes τ_{cp} and ς_{cp} is found in Section 6.5.

Table 1: Default parameter settings.

Parameter	Description	Default Value
maxiter	Maximum number of iterations	500
gamma11	Regularization weight for ℓ_1 model	0.006
gamma12	Regularization weight for ℓ_2 model	0.012
tprimldr	Step-size for primal DR	0.25
rhoprimaldr	Relaxation parameter for primal DR	1.25
rhoadmm	Penalty parameter for ADMM	1
tcp	Primal step size for Chambolle-Pock	$1/\ A\ _2$
scp	Dual step size for Chambolle-Pock	$1/\ A\ _2$

6.1 Effect of γ

The parameter $\gamma > 0$ controls the strength of the regularization term and therefore determines the extent of denoising applied in the reconstruction process. In particular, larger values of γ place greater emphasis on the total variation penalty, which promotes smoother recovered images and suppresses noise more effectively. However, excessively large values of γ may also oversmooth important image features, such as edges and fine detail, thereby producing a reconstruction that appears overly blurred. Consequently, the choice of γ is crucial, as it governs the trade-off between fidelity to the observed blurred image and the amount of regularization imposed on the solution. A suitable value of γ should therefore balance noise removal with the preservation of sharp structural features in order to yield a high-quality reconstruction.

To evaluate this effect, reconstructions are computed for several values of γ , with the blur and noise kernels held fixed (Gaussian kernel 9×9 with $\sigma = 2$, salt-and-pepper noise $d = 0.2$). The quality of each reconstructed image is then assessed using the composite score introduced in Section 4. Additional results using a motion blur kernel can be found in Appendix G.1, (Figures 8,9,10,11). Note that the following results hold for both gaussian and motion kernels.

Figures 1 and 2 displays the results for Gaussian blur with salt-and-pepper noise under the ℓ_1 model (sample reconstructions for the ℓ_2 model are found in Appendix G.1 Figure 12). As hypothesized, the regularization parameter γ controls the trade-off between denoising and deblurring. Across the blur and noise kernels considered in these experiments, larger values of γ favor denoising more strongly, leading to reconstructions with minimal visible noise but increased blur. When γ is chosen too large, the image becomes overly smooth and important structural details are lost. On the other hand, a value of γ around 0.01 appears to achieve the best compromise, removing a substantial amount of noise while still preserving the main edges and details of the image.

Figures 3, 4 illustrate the sensitivity of each algorithm to the regularization parameter γ . Figure 4 shows that PSNR and SSIM respond differently across the range of γ . In particular, PSNR tends to attain higher values at relatively smaller γ , where less aggressive regularization preserves pixel-wise fidelity, while SSIM peaks at moderately larger γ , where noise suppression improves structural consistency. Thus, PSNR and SSIM emphasize different aspects of reconstruction quality - PSNR favors low pixel-wise error (often at the expense of residual noise), whereas SSIM prioritizes structural coherence (often at the expense of slight smoothing). For the PSNR plot, it is also clear that with the exception of ADMM, all other algorithms have a flat fall-off and barely peak. Consequently, neither metric alone fully captures the trade-off between denoising and deblurring.

However, the composite transport distance in Figure 3 exhibits a more balanced behavior. The curves display a clear peak in the interior of the γ range and taper off toward both extremes, indicating that both insufficient regularization (noisy reconstructions) and excessive regularization (over-smoothed reconstructions) are penalized. This displays approximately symmetric concavity and provides

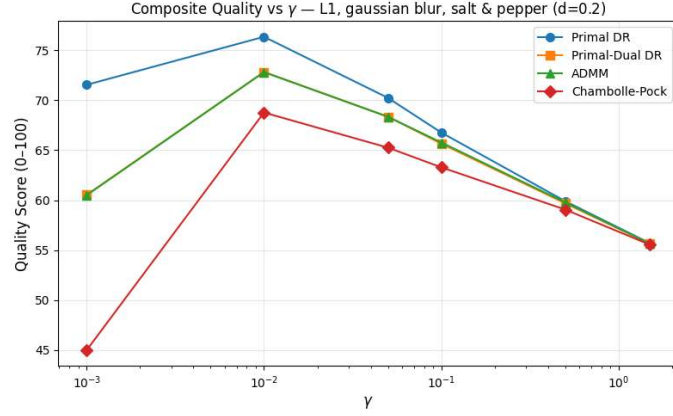


Figure 1: Composite quality score as a function of γ for all four algorithms.



Figure 2: Reconstructed images for each algorithm across selected values of γ using ℓ_1 regularization. Quality scores (Q, out of 100) are shown beneath each reconstruction.

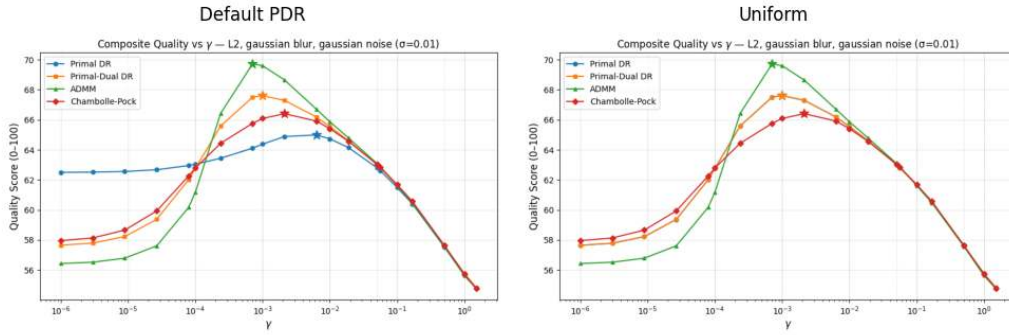


Figure 3: Gamma variations for the ℓ_2 problem, where the one on the right uses the same t, ρ for all algorithms and the one on the left uses the default parameters for PDR

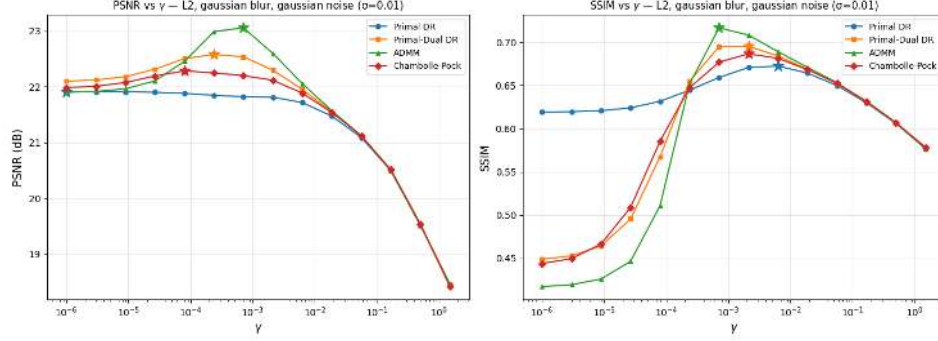


Figure 4: PSNR and SSIM variation with respect to γ

evidence of an optimal intermediate regime, where the trade-off between denoising and deblurring is best achieved.

The impact of the penalty parameter γ differs between the Douglas-Rachford style and Chambolle-Pock. In CP, although γ plays a role in the proximal of the duals, the algorithmic parameters t, s are decoupled from its effect. In the other algorithms γ plays a bigger role in how the choice of γ affects the parameters. In ADMM this becomes particularly important as γ indirectly controls the dual variable damping in addition to its usual role.

Table 2: The Core Structural Difference

	γ role in prox	γ role in linear solve	γ -parameter coupling
CP	Scales TV threshold directly	None	None
PDR	Scales TV threshold	None	Weak
PDDR	Scales TV threshold	Indirect via t	Moderate
ADMM	Scales TV threshold + damps dual	None	Strong

6.2 Golden section search for optimal γ

Since each evaluation of $\text{PSNR}(\gamma)$ requires running a full solver for a given algorithm to convergence, a dense grid search over γ is computationally prohibitive. Instead, we exploit the empirically observed unimodality of PSNR as a function of γ : as γ increases from zero, the TV penalty suppresses noise and PSNR rises beyond a problem dependent optimum, oversmoothing destroys edge structure and PSNR falls. This unimodal structure permits the use of golden section search [4], which is minimax-optimal among derivative-free bracketing methods for unimodal functions [7].

6.2.1 Method

Given a search interval $[a, b]$, the method places two interior probes at

$$c = b - \phi(b - a), \quad d = a + \phi(b - a),$$

where $\phi = (\sqrt{5} - 1)/2 \approx 0.618$ is the golden ratio conjugate. After comparing $f(c)$ and $f(d)$, the worse subinterval is discarded and exactly one new probe is placed in the retained subinterval, reusing the surviving evaluation from the previous step. Each iteration therefore reduces the interval length by a factor of ϕ , yielding $O(\log_{1/\phi}(\text{range}/\text{tol}))$ total solver calls. Since γ spans several orders of magnitude, the search is performed in $\log \gamma$ space over $\gamma \in [10^{-4}, 2]$ with a log-space tolerance of 0.05, requiring at most $\lceil \log_{1/\phi}(8.5/0.05) \rceil = 14$ solver runs per combination.

6.2.2 Results

The search was conducted under PDDR on a 256×256 greyscale image, across noise types (Gaussian, salt-and-pepper), data fidelity models (ℓ_1, ℓ_2), and noise levels $\sigma \in \{0.01, 0.05, 0.15, 0.30, 0.50\}$. Two trends emerge consistently:

- **γ^* increases monotonically with noise level.** Stronger corruption demands heavier TV regularization; e.g. under the ℓ_1 model with Gaussian noise, γ^* rises from 0.035 at $\sigma = 0.01$ to 0.540 at $\sigma = 0.50$.
- **Salt-and-pepper noise admits a much smaller γ^* than Gaussian.** The ℓ_1 fidelity directly absorbs impulse corruption, leaving little work for the TV term; e.g. $\gamma_{\ell_1}^* \approx 0.006$ at $\sigma = 0.05$ for S&P versus 0.165 for Gaussian at the same level.

Full results including the Huber model are in Appendix G.2; see also Figures 13 and 14 therein.

6.3 Regularization Strength vs. Image Resolution

We investigate whether γ^* , calibrated on a small image, transfers to higher resolutions. The golden section search of Section 6.2 was repeated across four scaled versions of the same scene (250 px to 1920 px wide) for all combinations of noise type, noise level, and model. Full figures are in Appendix G.3.

The central finding is that γ^* is *not* scale-invariant. For Gaussian noise under the ℓ_2 model, a log-log fit gives $\gamma^* \propto N^{0.51}$, meaning the optimal weight scales approximately as the square root of the total pixel count N . A γ calibrated at 250 px must thus be scaled up roughly $8\times$ when applied at 1920 px. In contrast, salt-and-pepper noise under the ℓ_1 model is nearly scale-invariant ($\gamma^* \propto N^{0.05}$): because the ℓ_1 fidelity handles impulse corruption directly, the TV weight remains nearly constant across resolutions. Best-case PSNR also improves consistently with resolution across all condition, so PSNR value from small test images should not be extrapolated to higher resolutions.

6.4 Kernel Misspecification

Since the true blurring kernel is unknown in practice, it is of interest to assess algorithm performance under kernel misspecification.

In this experiment, to evaluate the sensitivity of each algorithm to kernel misspecification, the image is first corrupted using a known blur kernel and salt-and-pepper noise, and then reconstructed using increasingly misspecified kernels. Four scenarios are considered. For Gaussian blur, the true kernel is fixed while either the standard deviation σ or the kernel size used for reconstruction is varied. For motion blur, the true kernel is fixed while either the angle or the length is varied. In each case, the first reconstruction kernel matches the true kernel exactly, while the remaining kernels introduce progressively larger levels of misspecification. This setup reflects a more realistic setting, in which the general type of blur may be identified visually, but its precise parameters remain unknown. Reconstruction quality is measured using the composite score described in Section 4. Figure 5 presents the results as grouped bar plots for all four algorithms.

The results paint a clear picture. In the Gaussian blur setting, reconstruction quality drops sharply as σ moves farther from its true value. Moreover, overestimating σ appears to be more harmful than underestimating it, with the score falling from about 83 to roughly 32 as σ increases from the true value of 2 to 5. By contrast, varying the Gaussian kernel size has almost no effect on reconstruction quality, suggesting that Gaussian blur is relatively robust to errors in kernel size.

Under motion blur, reconstruction quality is highly sensitive to errors in both the angle and the length of the kernel. Even an angular error of 15° or a small change in length leads to a decline in performance, with scores dropping from roughly 80 to below 40.

We note that, all four algorithms degrade at similar rates across all scenarios, suggesting that sensitivity to kernel mismatch is determined more by the underlying problem formulation than by the particular algorithm used.

6.5 Chambolle-Pock step-size

The Chambolle-Pock algorithm converges when the primal and dual step-sizes t and s satisfy $ts\|A\|_2^2$ (see Appendix D.1 for a discussion). The symmetric choice $t^* = s^* = \|A\|_2^{-1}$ saturates this bound in the sense that $t^*s^*\|A\|_2^2 = 1$, so that (t^*, s^*) is the largest symmetric step-size pair on the convergence boundary. Here, we study the impact that step-size selection has on Chambolle-Pock’s performance.

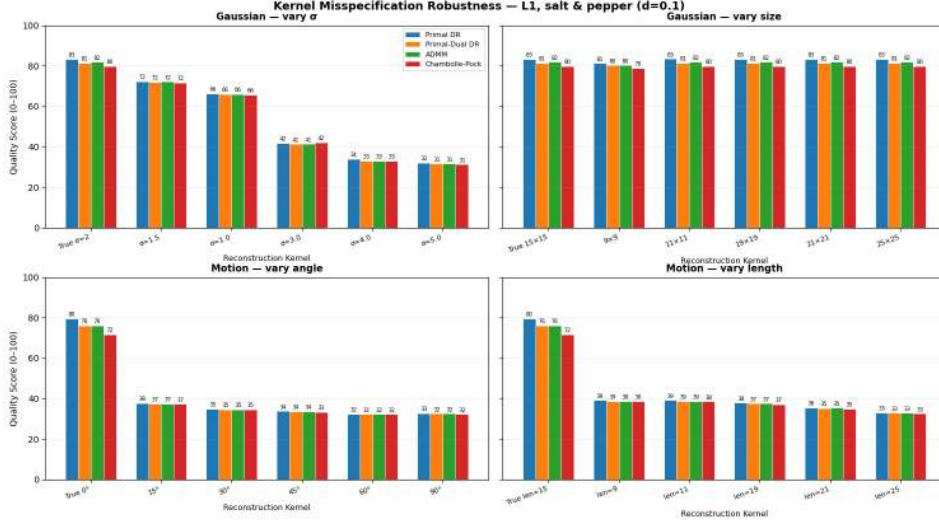


Figure 5: Reconstruction quality under kernel misspecification. Top left: Gaussian blur with varying σ . Top right: Gaussian blur with varying kernel size. Bottom left: motion blur with varying angle. Bottom right: motion blur with varying length. The first bar group in each panel corresponds to the correctly specified kernel.

Setup. We assess reconstruction quality under a 5×5 grid of step-sizes (t, s) . Both t and s are parameterized as multiples of the saturating step size: $t = m_t \cdot t^*$ and $s = m_s \cdot t^*$, with exponentially-spaced multipliers $m_t, m_s \in \{10^{-1}, 10^{-0.5}, 10^0, 10^{0.5}, 10^1\}$. The saturating pair (t^*, s^*) corresponds to the center of the grid at $(m_t, m_s) = (1, 1)$, while cells satisfying $m_t m_s < 1$ lie strictly inside the convergence region, and cells satisfying $m_t m_s > 1$ are not guaranteed to converge. We evaluate both ℓ_1 and ℓ_2 models under two blur kernels (15×15 Gaussian with $\sigma = 15$, and a length-15 motion blur with a 30° angle) and two noise regimes (salt-and-pepper with density $d = 0.2$, and Gaussian noise with $\sigma = 0.05$). Reconstruction quality is reported using the composite score defined in Section 4. A penalty weight of $\gamma = 0.05$ was used for both ℓ_1 and ℓ_2 models. Examples images used for this experiment are in Figure 18 in Appendix G.4.

Results. Figure 6 summarizes the results of this experiment on the cameraman test image when corrupted with Gaussian blurring + salt-and-pepper noise under the ℓ_1 model (top-left and bottom-left panels). Notably, the best score occurs outside of the strict convergence region: cells corresponding to step-sizes $t \in \{0.32t^*, 1.0t^*\}$ and $s \in \{3.16t^*, 10.0t^*\}$ outperform the saturating step-size pair (t^*, s^*) in the center of each grid. However, qualitatively, the reconstructions for these particular step-sizes are highly similar to one another (and in particular the reconstruction at (t^*, s^*)). This suggests that the additional quantitative performance achieved by step-sizes outside of the convergence region may not be worth the risk of catastrophic failure. Indeed, catastrophic failure (fully black reconstruction) occurs at the most extreme corner $t \in \{3.16t^*, 10.0t^*\}$ and $s \in \{3.16t^*, 10t^*\}$. From this, one concludes that the convergence condition $ts\|A\|^2 \geq 1$ is a conservative sufficient bound rather than a tight practical threshold. A secondary observation is the clear asymmetry in the effect of t and s on recovery: when the primal step-size is small $t \leq t^*$ one finds that increasing the dual step-size $s > s^*$ will improve quality (albeit modestly, with no convergence guarantee), while enlarging the primal step-size $t > t^*$ universally degrades reconstructions. The results for the ℓ_2 model are in the top-right and bottom-right panels of Figure 6. It is clear that the ℓ_2 model suffers from universally poor performance (quantitative and qualitative) for any choice of step-size, suggesting that the model appropriate for this noise regime.

Appendix G.4 presents the complete results of this experiment under the different permutations of blurring kernels (Gaussian, motion) and noise regimes (salt-and-pepper, Gaussian), yielding four different corruption regimes. Figure 19 and 21 presents the cameraman and manWithHat recovery score heatmaps for Gaussian kernels, while Figures 20 and 22 presents the reconstructions. Motion blur scores are in Figures 23 and 25, and reconstructions in Figures 24 and 26.

Of particular note is the poor performance of the ℓ_1 model under motion blurring, with only a modest gain over the ℓ_2 model when salt-and-pepper noise is applied. This suggests that the inherent

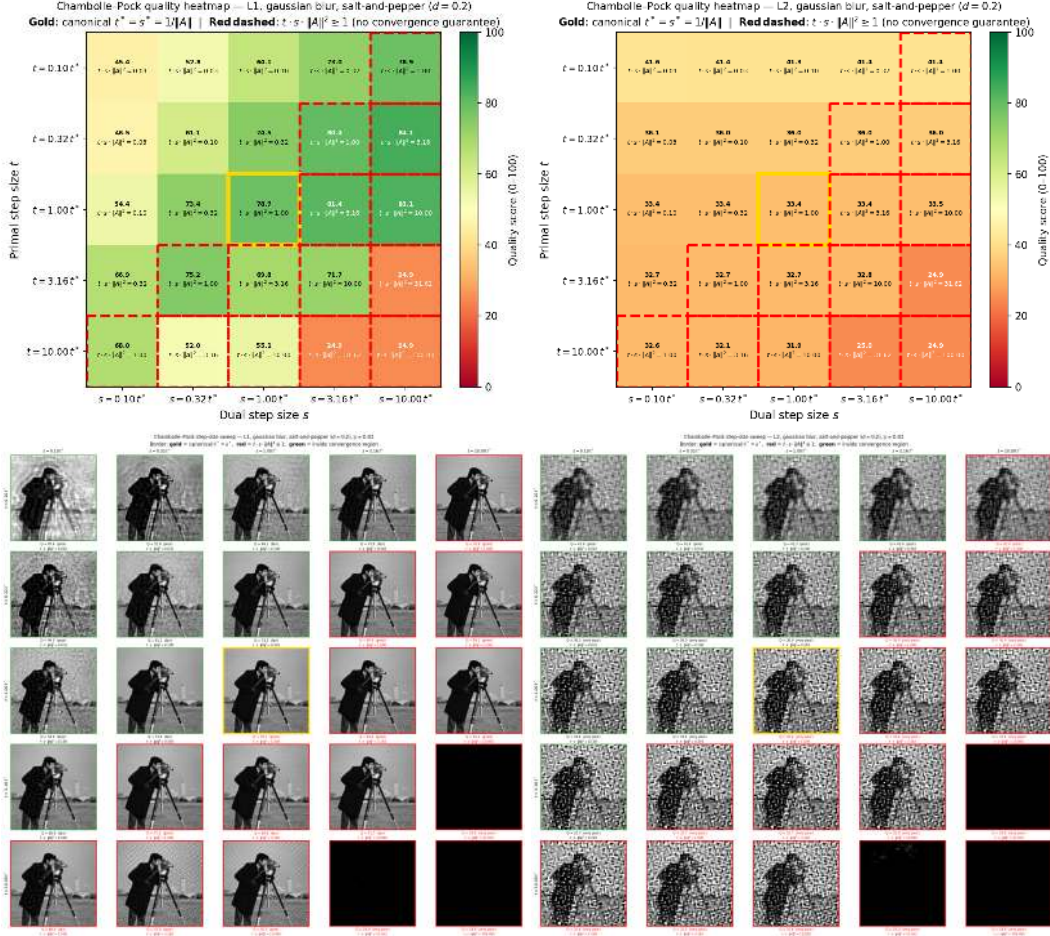


Figure 6: Chambolle-Pock quality scores (top) and reconstructions (bottom) on the cameraman image over a step-size grid under the ℓ_1 model (top-left, bottom-left) and ℓ_2 model (top-right, bottom-right). Gaussian blurring and salt-and-pepper noise. Cells outlined in red indicate when the step-size is outside the convergence region $ts\|A\| < 1$, with $t^* = s^* = \|A\|$ outlined in gold and in the center of each grid.

asymmetry caused by motion blur is more challenging than Gaussian blurring. Meanwhile, when Gaussian noise is applied on top of either blurring regime one finds that the ℓ_2 model outperforms the ℓ_1 model. We discuss the connection between model selection and noise regime in the following section.

Based on these results, a reasonable step-size heuristic for Chambolle-Pock is $t = s = \|A\|^{-1}$. As such, Chambolle-Pock is otherwise free of algorithm-specific tuning parameter tuning (the model parameter/penalty weight γ notwithstanding), positioning this algorithm as an attractive choice for non-expert practitioners who seek a simple off-the-shelf solution and do not wish to interact with nuanced parameter tuning.

6.6 Step Size and Relaxation Parameter Tuning for Douglas Rachford and ADMM Algorithms

To study the sensitivity of each algorithm to its tuning parameters, we perform a grid search over the step size $t \in \{0.25, 0.5, 0.75, 1.0, 1.5\}$ and relaxation parameter $\rho \in \{0.25, 0.5, 0.75, 1.0, 1.5\}$ for the three algorithms that use these parameters: Primal DR, Primal-Dual DR, and ADMM. We fix a motion blur kernel (length 40, angle 40°) with Gaussian noise ($\sigma = 0.1$) and use the ℓ_2 formulation with $\gamma = 0.01$ and a maximum of 1000 iterations. For each (t, ρ) combination, we record the composite quality score, PSNR, SSIM, and number of iterations to convergence. Results are presented in Appendix G.5,

Figures 27 and 28 reveal that for Primal DR and Primal-Dual DR, smaller values of t and ρ consistently produce better reconstructions, with the best quality achieved at $t = 0.25$ and $\rho = 0.25$. Both algorithms converge comfortably within 160 iterations across the entire grid, suggesting that while parameter choice affects solution quality, it does not threaten convergence. ADMM, by contrast, is far more sensitive: its quality scores remain flat at $Q \approx 39$ regardless of parameters, and the iteration heatmap shows that it fails to converge within 1000 iterations for most combinations. This suggests that ADMM requires substantially more careful tuning or more iterations to be competitive on this problem.

Figure 7 complements the heatmap analysis by examining how t and ρ interact with the regularization parameter γ . Using a Gaussian blur kernel with additive Gaussian noise ($\sigma = 0.01$) and the ℓ_2 formulation, we sweep t and ρ individually while fixing the other, and repeat this for three values of γ spanning several orders of magnitude: $\gamma = 10^{-6}$ (near-zero regularization), $\gamma = 0.005$ (moderate), and $\gamma = 0.99$ (heavy regularization). This reveals whether the sensitivity to step size and relaxation depends on the regularization regime.

The results confirm that parameter sensitivity is strongly modulated by γ . At small γ , the algorithms respond very differently to t and ρ : Primal DR favors small t , Primal Dual DR, is fairly robust to the choice of t , while ADMM favors large t . Furthermore, ADMM's quality collapses entirely for $\rho > 1.4$, while the 2 DR methods remain fairly stable across the full range of ρ . At moderate γ , all three algorithms become more stable, with quality varying by only about 1-2 points across the full range of t . This stability also follows when we vary ρ , apart from a sharp drop in quality in ADMM for large ρ . At large γ , the regularization term overwhelms the objective and the differences between algorithms and parameter choices is much less pronounced. This suggests that careful tuning of t and ρ is most important when γ is small, and that ADMM requires particular caution with the relaxation parameter ρ to avoid divergence. We also note that the behavior of ρ is almost identical for the DR methods, with the curves achieving nearly perfect overlap.

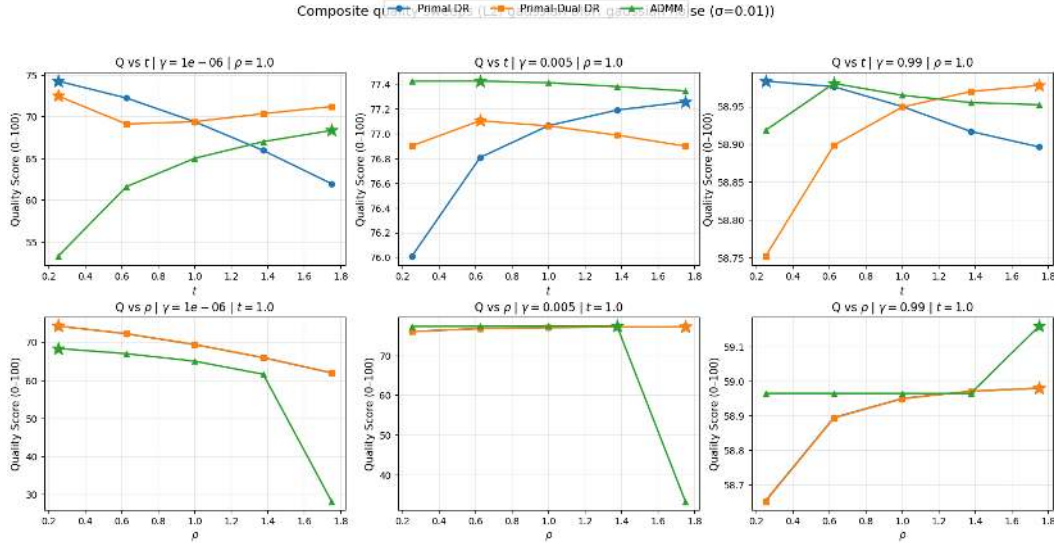


Figure 7: Variation of Douglas-Rachford Algorithms with respect to step sizes t, ρ for 3 γ points selected across the entire interval

6.7 Remarks

Throughout our experiments, we observe a clear relationship between the tested noise regime (salt-and-pepper/Gaussian) and model choice (ℓ_1/ℓ_2). In particular, the ℓ_1 model is generally favored under the sparse, high-impulse residuals induced by salt-and-pepper noise. Meanwhile, the ℓ_2 model (inherently nonsparse due to the ℓ_2 -norm's smoothness) consistently fails to handle the sparse residuals caused by salt-and-pepper noise, and no choice of algorithm/algorithm parameters can compensate for this. This hypothesis is supported by the results under Gaussian noise, wherein the residuals are nonsparse. Indeed, under Gaussian noise one finds the nonsparse ℓ_2 model to

outperform the sparse ℓ_1 model. This connection emphasizes the importance that model selection has on performance, and practitioners should be careful to select a model that corresponds to the modality of corruption.

7 Critique and Future Directions

7.1 Critique

- **Downsides of FFT** - FFT assumes periodic boundary conditions which may, introduce artifacts of circular convolution. In standard digital signal processing this is typically alluded using pixel-cushioning, e.g. zero-padding.
- **Role of γ** - γ has the same broad effect on all algorithms in that it controls a trade-off between data fidelity and smoothness, that is, between deblurring and denoising. However, it influences the internal structure of each algorithm differently. As such, a standardized comparison of all algorithms on the basis of γ will likely not produce deterministic conclusions. This is particularly true for Chambolle-Pock and the Douglas-Rachford splitting algorithms (including ADMM).
- **Metric Selection** - PSNR and SSIM are well established quality metrics and the transport distance is complementary to their shortcomings. The latter, however, does not satisfy all the conditions of a *true* metric. It has been designed in a heuristic fashion. Moreover, the problem has been formulated using the total variation metric with the ℓ_1, ℓ_2 norms - the evaluation metrics used in the experiments are not necessarily well-aligned. That is, there may be a subjective discrepancy between visual perception and the information conveyed using composite metric, although both capture the notion of quality (just not in the same way).¹
- **ℓ_1 convergence under default settings** - Under the shipped defaults, all four algorithms exhaust the 500-iteration cap on the ℓ_1 problem while the ℓ_2 problem converges in < 200 iterations; see Appendix F for details and mitigations.

7.2 Future Directions

This section has been written from the perspective of future directions when it comes to image deblurring and denoising and not better ways to formulate it as a convex optimization objective.

Analytic Methods - The original problem has two distinct corruption processes - blurring and noising. However, the vast majority of prior work considers each separately. Anisotropic diffusion [8] has been proven to be exceptionally capable in denoising whereas more sophisticated physics-based approaches such as tomography has been admitted in a variety of tasks particularly in non-destructive testing. The other alternative for this will be to use multi-spectral (wavelet) analysis [1]

Machine Learning and Probabilistic Methods, Sequential Control - The forward blur operator K can be interpreted as a non-injective mapping that induces a structural collapse of the signal space. In particular, multiple images $x_1 \neq x_2$ may produce identical observations $Kx_1 = Kx_2$, implying that the forward model defines an equivalence relation over $\mathbb{R}^n$ and therefore acts as a quotient map rather than an invertible transformation. From this perspective, image degradation is more naturally viewed as a homomorphic mapping that preserves certain coarse geometric structures (e.g., low-frequency content and global intensity patterns) while irreversibly discarding high-frequency information. There is potential for upcoming work that

- provides a proof that the blurring kernel K is a contraction in some probability metric;
- uses optimal transport for providing guidance on the approximation of inverse mappings in lieu of the recent advances in representation learning and algebraic equivalence in controlled Markov chains [9].

¹The transport distance does not exactly match the optimization formulation, but *is* aligned to the underlying image deblurring and denoising task.

8 Individual Contributions

All members of the group have contributed to the following parts:

- Mathematical formulation of the problem: calculation of proximal mappings, operator definitions, etc.
- The main scientific study: the analysis of parameter sensitivity with respect to γ, t, ρ for all algorithms.

In addition, each member has contributed individually to the following:

- Daniel Bensimon (260988886). Worked on the Primal Douglas Rachford algorithm (Section 3.2). Worked on the implementation of the algorithm. Worked on the proximal operators that were used in the final package (Although all group members independently implemented their own proximal operators). Designed and implemented the kernel misspecification and gamma varying experiments found in Sections 6.1, 6.4.
- David Fleischer (260396047). Worked on the introduction & problem formulation (Section 1), proximal operator summaries (Sections 2 and 3.1). Worked on the Chambolle-Pock derivation/convergence analysis (Section 3.5 and Appendix D.1). Worked on the Chambolle-Pock implementation and steps-size experiments (Section 6.5 and Appendix G.4), and the final experimental remarks in Section 6.7.
- Arka Ian Goswami (261258780). Independently worked on problem formulation, provided the general background on proximal, splitting methods, constrained optimization (Appendices A, B), and suggested an adaptive corruption pipeline with regards to image resolution as has been implemented by Aariyan in the package, Implemented the ADMM algorithm (Section 3.4), attempted a convergence proof (Appendix C) using Lyapunov descent, conducted experiments over γ, t, ρ for the ℓ_2 problem, displayed in figures 3, 4, 12.
Responsible for quality metrics (Section 4) and designed the heuristic composite metric. **Note** - Aariyan came up with the PSNR metric independently too.
Wrote the future directions and critique sections (Section 7) - identification of the varying influence of γ was also implicitly covered, by myself 6.1 and independently, by Daniel Bensimon at the beginning of 6.1.
- Aariyan al Mamum (261180608). Worked on the Primal-Dual Douglas Rachford algorithm (Section 3) derivation and its implementation. Built a corruption pipeline covering FFT-based blurring and a range of noise types, including Gaussian and salt-and-pepper noise and multilayered noise. Designed and implemented the Golden Section Search procedure for optimal γ (Section 6.2). Conducted the γ vs image resolution study (Section 6.3), establishing scaling laws for the regularization parameter across different image sizes and noise regimes.

Regardless of each member's individual contribution, all sections and material in the main body of the document has been collectively agreed upon and everyone is sufficiently-versed in all aspects of the project. No part was done individually or in isolation.

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A General Splitting Algorithms and Proximal Point Methods

A.1 Proximal Point Method

The proximal point method is an idea that converts a possibly ill-posed, at least from a computability perspective, problem into a well-posed one. The reference for the majority of this section is **Parikh and Boyd**, and **Constantine Caramanis**.

A.2 Proximal Point Method and the Resolvent Operator

Consider the following optimization problem posed as a root-finding problem

$$0 \in F(x),$$

where F is a maximal monotone operator (for example, a subdifferential of a convex function). Directly solving this inclusion is often difficult. The proximal point method (PPM) addresses this by introducing a regularized problem at each iteration. Given an iterate x^k , the proximal point method defines x^{k+1} implicitly as the solution to

$$0 \in F(x^{k+1}) + \frac{1}{t}(x^{k+1} - x^k),$$

where $t > 0$ is a stepsize parameter. This modifies the original problem by adding a strongly monotone term $\frac{1}{t}I$, which ensures improved stability and uniqueness of the solution. Rearranging the inclusion, we obtain

$$0 \in tF(x^{k+1}) + x^{k+1} - x^k \implies x^k \in x^{k+1} + tF(x^{k+1}) \implies x^k \in (I + tF)(x^{k+1}).$$

Applying the inverse operator $(I + tF)^{-1}$ to both sides yields

$$x^{k+1} = (I + tF)^{-1}(x^k).$$

The mapping

$$R_{tF} := (I + tF)^{-1}$$

is called the *resolvent* of F . Thus, the proximal point iteration can be written compactly as

$$x^{k+1} = R_{tF}(x^k).$$

A point x^* solves the inclusion $0 \in F(x^*)$ if and only if it is a fixed point of the resolvent

$$x^* = R_{tF}(x^*).$$

Therefore, the proximal point method reduces the problem of finding a root of F to finding a fixed-point of the resolvent. This makes the overall problem significantly easier due to the **firm non-expansiveness of the resolvent**.

If $F = \partial f$ for a closed, convex function f , then the resolvent corresponds to the proximal operator

$$R_{t\partial f}(x^k) = \arg \min_x \left[f(x) + \frac{1}{2t} \|x - x^k\|_2^2 \right].$$

Hence, the proximal point method generalizes proximal optimization to the broader setting of monotone operator theory. While the resolvent $R_{tF} = (I + tF)^{-1}$ provides a convergent fixed-point iteration, it is often computationally intractable when $F = A + B$. In such cases, one cannot efficiently compute $(I + t(A + B))^{-1}$ directly. This motivates operator splitting methods, such as Douglas–Rachford (all variations), which instead construct iterations using the individual resolvents $(I + tA)^{-1}$ and $(I + tB)^{-1}$.

A.3 Douglas–Rachford Splitting

Using earlier notation, $R_{(\cdot)}$ denotes the resolvent and $C_{(\cdot)}$ the Cayley operator:

$$R_A = (I + tA)^{-1}, \quad C_A = 2R_A - I = 2(I + tA)^{-1} - I.$$

Proposition A.1. *If $0 \in A(x) + B(x)$ (i.e., x is a solution), then the following are equivalent:*

$$\begin{aligned} C_A \circ C_B(z) &= z \quad (\text{fixed point}), \\ x &= R_B(z). \end{aligned}$$

Consider $0 \in A(x) + B(x)$, where A arises from proximal components (subdifferentials) and B encodes linear coupling. A natural iteration is

$$z_{t+1} = C_A \circ C_B(z_t),$$

the Peaceman–Rachford step. Douglas–Rachford instead averages with the identity

$$z_{t+1} = (\lambda I + (1 - \lambda)C_A \circ C_B)z_t,$$

with classical choice $\lambda = \frac{1}{2}$,

$$z_{t+1} = \frac{1}{2}(I + C_A \circ C_B)z_t.$$

Under maximal monotonicity and existence of a solution, this converges to a fixed point.

A.4 Douglas–Rachford Iteration

One may alternate the variable updates and each alternate method would yield a new set of equivalent updates. One such valid iteration is the following -

Let $t > 0$ and define $R_A = (I + tA)^{-1}$, $R_B = (I + tB)^{-1}$. Starting from z^0 ,

$$\begin{aligned} x^k &= R_A(z^{k-1}), \\ y^k &= R_B(2x^k - z^{k-1}), \\ z^k &= z^{k-1} + \rho(y^k - x^k), \end{aligned}$$

with $\rho \in (0, 2)$. Equivalently, $z^k = F(z^{k-1})$ where

$$F(z) = z + (I + tB)^{-1}(2(I + tA)^{-1}(z) - z) - (I + tA)^{-1}(z).$$

If A, B are maximal monotone and the solution set is nonempty, then $z^k \rightarrow z^*$ and

$$x^* = R_B(z^*).$$

B Overview of Optimization Under Constrained Objectives

First, one should account for the simple primal formulation below

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) \\ \text{subject to} \quad & h_i(x) = 0, \quad i = 1, \dots, m, \\ & g_j(x) \leq 0, \quad j = 1, \dots, p, \end{aligned}$$

where f , h_i , and g_j are convex functions. Hence, here one has an objective function and equality and inequality constraints. This is the typical form of a linear program (LP). A common, and elegant, way to consume the significance of the constraints and produce an unconstrained objective is using *Lagrangian Relaxations* [Rockafellar, Bertsekas, Bertsekas]

$$\mathcal{L}(x, \lambda, \nu) = f(x) + \sum_{i=1}^m \lambda_i h_i(x) + \sum_{j=1}^p \nu_j g_j(x),$$

where λ, ν are the *Lagrange Multipliers* for the equality and inequality constraints respectively. Carrying on, one may define the Lagrange dual function as

$$d(\lambda, \nu) = \inf_x \mathcal{L}(x, \lambda, \nu) \leq f(x^*)$$

which satisfies the Fenchel-Rockafellar duality conditions.

B.1 Dual

The dual formulation for the aforementioned primal problem is - $\max_{\lambda, \nu \geq 0} d(\lambda, \nu)$ which is always a concave maximization problem, even when the primal is not strictly convex. The characterizations of the duality strength are **weak duality** - for any feasible x and $(\lambda, \nu \geq 0)$,

$$d(\lambda, \nu) \leq f(x).$$

and **strong duality** - if the primal version is convex and satisfies a constraint qualification [Slater]

$$\max_{\lambda, \nu \geq 0} d(\lambda, \nu) = \min_x f(x).$$

B.2 Karush–Kuhn–Tucker (KKT) Conditions

The KKT conditions are a series of necessary and sufficient properties that a triple (x^*, λ^*, ν^*) need to satisfy to be identified as a solution to the dual problem. Under strong duality, a point (x^*, λ^*, ν^*) is optimal if and only if it satisfies

Primal feasibility:

$$\begin{aligned} h_i(x^*) &= 0, \quad i = 1, \dots, m, \\ g_j(x^*) &\leq 0, \quad j = 1, \dots, p. \end{aligned}$$

Dual feasibility:

$$\nu_j^* \geq 0, \quad j = 1, \dots, p.$$

Stationarity:

$$0 \in \partial f(x^*) + \sum_{i=1}^m \lambda_i^* \partial h_i(x^*) + \sum_{j=1}^p \nu_j^* \partial g_j(x^*).$$

If all functions are differentiable, this reduces to

$$\nabla f(x^*) + \sum_{i=1}^m \lambda_i^* \nabla h_i(x^*) + \sum_{j=1}^p \nu_j^* \nabla g_j(x^*) = 0.$$

Complementary Slackness:

$$\nu_j^* g_j(x^*) = 0, \quad j = 1, \dots, p.$$

C Convergence of ADMM

The convergence proofs follow two steps - one, to establish that the iterative process produces a monotonically decreasing sequence of residuals (Cauchyness), and another for convergence to the optimal solution (fixed-point). First consider the update

$$x^k = (I + A^\top A)^{-1} \left(u^{k-1} + A^\top y^{k-1} - \frac{1}{t}(w^{k-1} + A^\top z^{k-1}) \right), \quad (17)$$

$$(18)$$

from this step one may infer the following

$$x^{k+1} = (I + A^\top A)^{-1} \left(u^k + A^\top y^k - \frac{1}{t}(w^k + A^\top z^k) \right), \quad (19)$$

$$x^k = (I + A^\top A)^{-1} \left(u^{k-1} + A^\top y^{k-1} - \frac{1}{t}(w^{k-1} + A^\top z^{k-1}) \right). \quad (20)$$

Subtracting the second relation from the first gives

$$x^{k+1} - x^k = (I + A^\top A)^{-1} \left[u^k + A^\top y^k - \frac{1}{t}(w^k + A^\top z^k) - \left(u^{k-1} + A^\top y^{k-1} - \frac{1}{t}(w^{k-1} + A^\top z^{k-1}) \right) \right] \quad (21)$$

$$= (I + A^\top A)^{-1} \left[u^k - u^{k-1} + A^\top (y^k - y^{k-1}) - \frac{1}{t}(w^k - w^{k-1}) - \frac{1}{t}A^\top (z^k - z^{k-1}) \right]. \quad (22)$$

Taking norms on both sides, we obtain

$$\|x^{k+1} - x^k\| = \left\| (I + A^\top A)^{-1} \left[u^k - u^{k-1} + A^\top (y^k - y^{k-1}) - \frac{1}{t}(w^k - w^{k-1}) - \frac{1}{t}A^\top (z^k - z^{k-1}) \right] \right\| \quad (23)$$

$$\leq \|(I + A^\top A)^{-1}\| \left\| u^k - u^{k-1} + A^\top (y^k - y^{k-1}) - \frac{1}{t}(w^k - w^{k-1}) - \frac{1}{t}A^\top (z^k - z^{k-1}) \right\|. \quad (24)$$

Applying the triangle inequality yields

$$\|x^{k+1} - x^k\| \leq \|(I + A^\top A)^{-1}\| \left(\|u^k - u^{k-1}\| + \|A^\top (y^k - y^{k-1})\| \right) \quad (25)$$

$$+ \frac{1}{t}\|w^k - w^{k-1}\| + \frac{1}{t}\|A^\top (z^k - z^{k-1})\| \Big). \quad (26)$$

submultiplicativity of the operator norm now implies

$$\|A^\top (y^k - y^{k-1})\| \leq \|A^\top\| \|y^k - y^{k-1}\|, \quad (27)$$

$$\|A^\top (z^k - z^{k-1})\| \leq \|A^\top\| \|z^k - z^{k-1}\|. \quad (28)$$

Therefore,

$$\|x^{k+1} - x^k\| \leq \|(I + A^\top A)^{-1}\| \left(\|u^k - u^{k-1}\| + \|A^\top\| \|y^k - y^{k-1}\| \right) \quad (29)$$

$$+ \frac{1}{t}\|w^k - w^{k-1}\| + \frac{1}{t}\|A^\top\| \|z^k - z^{k-1}\| \Big). \quad (30)$$

If, in addition, we use the fact that $A^\top A \succeq 0$, hence

$$\|(I + A^\top A)^{-1}\| \leq 1$$

in the Euclidean operator norm, then we obtain the simpler bound

$$\|x^{k+1} - x^k\| \leq \|u^k - u^{k-1}\| + \|A^\top\| \|y^k - y^{k-1}\| + \frac{1}{t} \|w^k - w^{k-1}\| + \frac{1}{t} \|A^\top\| \|z^k - z^{k-1}\|. \quad (31)$$

Thus, one may provide a convergence proof for the target variable x through the convergence of the other variables which entirely characterize x . However, they are actually coupled. Hence, not separable and individual contraction arguments for each normed-error will typically not be easy to provide.

C.1 Suboptimality Gaps and Fixed-Point Convergence Analysis

To establish convergence of the coupled variables, one needs to introduce a continuous mapping which depends on all coupled variables. This captures both primal feasibility and stabilization of the algorithm. The key to the proof is to identify an energy which decreases at every iteration. From the variable dependencies in this scenario the following suitable functional

$$\Phi^k := \frac{1}{t} \|w^k - w^*\|_2^2 + \frac{1}{t} \|z^k - z^*\|_2^2 + t \|u^k - u^*\|_2^2 + t \|y^k - y^*\|_2^2. \quad (32)$$

captures the abstract notion of energy in the image denoising and deblurring problem. The purpose of the proof is to show that Φ^k is nonincreasing, and that its decrease controls both the residuals and the successive differences of the primal blocks.

C.2 A one-step descent inequality

For every $k \geq 1$,

$$\frac{1}{t} \|w^k - w^*\|_2^2 = \frac{1}{t} \|w^{k-1} - w^*\|_2^2 + 2 \langle w^{k-1} - w^*, r_1^k \rangle + t \|r_1^k\|_2^2, \quad (33)$$

$$\frac{1}{t} \|z^k - z^*\|_2^2 = \frac{1}{t} \|z^{k-1} - z^*\|_2^2 + 2 \langle z^{k-1} - z^*, r_2^k \rangle + t \|r_2^k\|_2^2. \quad (34)$$

Proof. This is immediate from

$$w^k - w^* = (w^{k-1} - w^*) + tr_1^k, \quad z^k - z^* = (z^{k-1} - z^*) + tr_2^k, \quad (35)$$

followed by expansion of the squares. $\square$

C.3 Convexity bounds

For every $k \geq 1$,

$$f(u^k) - f(u^*) \leq \langle w^k, u^k - u^* \rangle, \quad (36)$$

$$g(y^k) - g(y^*) \leq \langle z^k, y^k - y^* \rangle, \quad (37)$$

$$f(u^k) - f(u^*) \geq \langle w^*, u^k - u^* \rangle, \quad (38)$$

$$g(y^k) - g(y^*) \geq \langle z^*, y^k - y^* \rangle. \quad (39)$$

Proof. Since $w^k \in \partial f(u^k)$, the subgradient inequality gives

$$f(u^*) \geq f(u^k) + \langle w^k, u^* - u^k \rangle, \quad (40)$$

which is equivalent to (36). Similarly, $z^k \in \partial g(y^k)$ gives (37).

Since $w^* \in \partial f(u^*)$, we also have

$$f(u^k) \geq f(u^*) + \langle w^*, u^k - u^* \rangle, \quad (41)$$

which is (38). The proof of (39) is identical. $\square$

C.4 One-step Lyapunov descent

For every $k \geq 1$,

$$\Phi^{k-1} - \Phi^k \geq t\|r_1^k\|_2^2 + t\|r_2^k\|_2^2 + t\|u^k - u^{k-1}\|_2^2 + t\|y^k - y^{k-1}\|_2^2. \quad (42)$$

Proof. Rearranging the square identities (33)–(34),

$$\frac{1}{t}\|w^{k-1} - w^*\|_2^2 - \frac{1}{t}\|w^k - w^*\|_2^2 = -2\langle w^{k-1} - w^*, r_1^k \rangle - t\|r_1^k\|_2^2, \quad (43)$$

$$\frac{1}{t}\|z^{k-1} - z^*\|_2^2 - \frac{1}{t}\|z^k - z^*\|_2^2 = -2\langle z^{k-1} - z^*, r_2^k \rangle - t\|r_2^k\|_2^2. \quad (44)$$

Next, expanding the primal parts

$$t\|u^{k-1} - u^*\|_2^2 - t\|u^k - u^*\|_2^2 = 2t\langle u^{k-1} - u^k, u^k - u^* \rangle + t\|u^{k-1} - u^k\|_2^2, \quad (45)$$

$$t\|y^{k-1} - y^*\|_2^2 - t\|y^k - y^*\|_2^2 = 2t\langle y^{k-1} - y^k, y^k - y^* \rangle + t\|y^{k-1} - y^k\|_2^2. \quad (46)$$

Now sum (43), (44), (45), and (46). We obtain

$$\begin{aligned} \Phi^{k-1} - \Phi^k &= -2\langle w^{k-1} - w^*, r_1^k \rangle - 2\langle z^{k-1} - z^*, r_2^k \rangle \\ &\quad + 2t\langle u^{k-1} - u^k, u^k - u^* \rangle + 2t\langle y^{k-1} - y^k, y^k - y^* \rangle \\ &\quad - t\|r_1^k\|_2^2 - t\|r_2^k\|_2^2 + t\|u^k - u^{k-1}\|_2^2 + t\|y^k - y^{k-1}\|_2^2. \end{aligned} \quad (47)$$

The inner-products may now be simplified by noting that

$$w^{k-1} = w^k - tr_1^k, \quad z^{k-1} = z^k - tr_2^k. \quad (48)$$

Hence

$$-2\langle w^{k-1} - w^*, r_1^k \rangle = -2\langle w^k - w^*, r_1^k \rangle + 2t\|r_1^k\|_2^2, \quad (49)$$

$$-2\langle z^{k-1} - z^*, r_2^k \rangle = -2\langle z^k - z^*, r_2^k \rangle + 2t\|r_2^k\|_2^2. \quad (50)$$

Substitute these into (47)

$$\begin{aligned} \Phi^{k-1} - \Phi^k &= -2\langle w^k - w^*, r_1^k \rangle - 2\langle z^k - z^*, r_2^k \rangle \\ &\quad + 2t\langle u^{k-1} - u^k, u^k - u^* \rangle + 2t\langle y^{k-1} - y^k, y^k - y^* \rangle \\ &\quad + t\|r_1^k\|_2^2 + t\|r_2^k\|_2^2 + t\|u^k - u^{k-1}\|_2^2 + t\|y^k - y^{k-1}\|_2^2. \end{aligned} \quad (51)$$

Now using

$$r_1^k = (x^k - u^k) = (x^k - u^*) - (u^k - u^*), \quad r_2^k = (Ax^k - y^k) = (Ax^k - y^*) - (y^k - y^*).$$

From the KKT relation $w^* + A^\top z^* = 0$, together with primal feasibility $u^* = x^*$, $y^* = Ax^*$. In fact,

$$-2\langle w^k - w^*, r_1^k \rangle - 2\langle z^k - z^*, r_2^k \rangle + 2t\langle u^{k-1} - u^k, u^k - u^* \rangle + 2t\langle y^{k-1} - y^k, y^k - y^* \rangle \quad (52)$$

$$= 2\langle w^k - w^*, u^k - u^* \rangle + 2\langle z^k - z^*, y^k - y^* \rangle - 2t\langle u^{k-1} - u^k, r_1^k \rangle - 2t\langle y^{k-1} - y^k, r_2^k \rangle. \quad (53)$$

By (36) and (38) (convexity),

$$\langle w^k - w^*, u^k - u^* \rangle \geq 0. \quad (54)$$

Likewise, (37) and (39) imply

$$\langle z^k - z^*, y^k - y^* \rangle \geq 0. \quad (55)$$

We now estimate the remaining mixed parts in (53) using Young's inequality

$$2|\langle a, b \rangle| \leq \|a\|^2 + \|b\|^2.$$

Applying this gives

$$-2t\langle u^{k-1} - u^k, r_1^k \rangle \geq -t\|u^{k-1} - u^k\|_2^2 - t\|r_1^k\|_2^2,$$

$$-2t\langle y^{k-1} - y^k, r_2^k \rangle \geq -t\|y^{k-1} - y^k\|_2^2 - t\|r_2^k\|_2^2.$$

Substituting these bounds into (51) yields

$$\Phi^{k-1} - \Phi^k \geq t\|r_1^k\|_2^2 + t\|r_2^k\|_2^2 + t\|u^k - u^{k-1}\|_2^2 + t\|y^k - y^{k-1}\|_2^2 \geq 0,$$

and hence

$$\Phi^{k-1} - \Phi^k \geq 0 \implies \Phi^{k-1} \geq \Phi^k,$$

which is exactly (42). $\square$

The inequalities (54)–(55) follow from monotonicity of the subdifferentials ∂f and ∂g , which holds under convexity of f and g . Under strong convexity, one can establish linear convergence rates (see, e.g., Nishihara). Hence, the Lyapunov functional is monotonically decreasing with respect to the iteration index.

C.5 Global convergence

Now one may establish this property throughout the entire iterative process as follows - Assume f and g are proper, closed, convex, and that the split problem admits a KKT point $(x^*, u^*, y^*, w^*, z^*)$. Then the ADMM iterates satisfy:

$$\begin{aligned} x^k - u^k &\rightarrow 0, \\ Ax^k - y^k &\rightarrow 0, \\ u^k - u^{k-1} &\rightarrow 0, \\ y^k - y^{k-1} &\rightarrow 0, \\ w^k + A^\top z^k &\rightarrow 0. \end{aligned}$$

Moreover, every cluster point of $\{(x^k, u^k, y^k, w^k, z^k)\}$ is a KKT point, hence is primal-dual optimal. If the KKT point is unique, then the full sequence converges to it.

Proof. Sum (42) from $k = 1$ to N :

$$\sum_{k=1}^N \left(t\|r_1^k\|_2^2 + t\|r_2^k\|_2^2 + t\|u^k - u^{k-1}\|_2^2 + t\|y^k - y^{k-1}\|_2^2 \right) \leq \Phi^0 - \Phi^N.$$

Since $\Phi^N \geq 0$,

$$\sum_{k=1}^{\infty} \left(\|r_1^k\|_2^2 + \|r_2^k\|_2^2 + \|u^k - u^{k-1}\|_2^2 + \|y^k - y^{k-1}\|_2^2 \right) < \infty.$$

Therefore,

$$r_1^k \rightarrow 0, \quad r_2^k \rightarrow 0, \quad u^k - u^{k-1} \rightarrow 0, \quad y^k - y^{k-1} \rightarrow 0.$$

Equivalently,

$$x^k - u^k \rightarrow 0, \quad Ax^k - y^k \rightarrow 0.$$

Now use the rewritten x -stationarity identity:

$$w^k + A^\top z^k = t(u^{k-1} - u^k) + tA^\top(y^{k-1} - y^k).$$

Since $u^{k-1} - u^k \rightarrow 0$ and $y^{k-1} - y^k \rightarrow 0$, it follows immediately that

$$w^k + A^\top z^k \rightarrow 0.$$

Now let $(\bar{x}, \bar{u}, \bar{y}, \bar{w}, \bar{z})$ be any cluster point of the sequence. Passing to a convergent subsequence if needed, the limits above give

$$\bar{x} - \bar{u} = 0, \quad A\bar{x} - \bar{y} = 0, \quad \bar{w} + A^\top \bar{z} = 0.$$

Furthermore, from

$$w^k \in \partial f(u^k), \quad z^k \in \partial g(y^k),$$

and the closedness of the subdifferential graph for proper lower semicontinuous convex functions, we obtain

$$\bar{w} \in \partial f(\bar{u}), \quad \bar{z} \in \partial g(\bar{y}).$$

Hence $(\bar{x}, \bar{u}, \bar{y}, \bar{w}, \bar{z})$ satisfies the full KKT system and is therefore optimal.

Finally, if the KKT point is unique, then every cluster point coincides with the same limit, so the whole sequence converges. $\square$

As the proof earlier might be a little long and as it might be difficult to get a broad picture, consider the summary below:

1. The relations

$$w^k \in \partial f(u^k), \quad z^k \in \partial g(y^k),$$

say that the dual variables are not arbitrary, they are precisely the subgradients induced by the proximal subproblems.

2. The residuals

$$r_1^k = x^k - u^k, \quad r_2^k = Ax^k - y^k,$$

measure violation of the two split constraints. Their convergence to zero is exactly primal feasibility.

3. The Lyapunov function

$$\Phi^k = \frac{1}{t} \|w^k - w^*\|_2^2 + \frac{1}{t} \|z^k - z^*\|_2^2 + t \|u^k - u^*\|_2^2 + t \|y^k - y^*\|_2^2$$

couples primal and dual errors in a way adapted to the ADMM updates.

4. The one-step estimate (42) is the analogue of a descent lemma. It shows that the decrease of Φ^k controls both infeasibility and lack of stabilization of the primal blocks.
5. Telescoping converts local one-step improvement into global convergence.

C.6 Residual Convergence

To establish convergence of the coupled variables, one needs to introduce a continuous mapping which depends on all coupled variables. This captures both primal feasibility and stabilization of the algorithm.

Incremental Lyapunov functional. Define

$$\Psi^{k+1} := t \|r_1^{k+1}\|_2^2 + t \|r_2^{k+1}\|_2^2 + t \|u^{k+1} - u^k\|_2^2 + t \|y^{k+1} - y^k\|_2^2,$$

where

$$r_1^{k+1} := x^{k+1} - u^{k+1}, \quad r_2^{k+1} := Ax^{k+1} - y^{k+1}.$$

This functional measures both constraint violation and lack of stabilization between consecutive iterates.

Relation to multiplier increments. From the dual updates,

$$w^{k+1} - w^k = t r_1^{k+1}, \quad z^{k+1} - z^k = t r_2^{k+1},$$

so that

$$\frac{1}{t} \|w^{k+1} - w^k\|_2^2 = t \|r_1^{k+1}\|_2^2, \quad \frac{1}{t} \|z^{k+1} - z^k\|_2^2 = t \|r_2^{k+1}\|_2^2.$$

Thus Ψ^{k+1} may equivalently be written as

$$\Psi^{k+1} = \frac{1}{t} \|w^{k+1} - w^k\|_2^2 + \frac{1}{t} \|z^{k+1} - z^k\|_2^2 + t \|u^{k+1} - u^k\|_2^2 + t \|y^{k+1} - y^k\|_2^2.$$

Summability and vanishing increments. From the descent inequality established earlier, we obtain

$$\sum_{k=0}^{\infty} \Psi^{k+1} < \infty.$$

Hence,

$$r_1^{k+1} \rightarrow 0, \quad r_2^{k+1} \rightarrow 0, \quad u^{k+1} - u^k \rightarrow 0, \quad y^{k+1} - y^k \rightarrow 0.$$

Propagation across variables. From the multiplier updates,

$$w^{k+1} - w^k \rightarrow 0, \quad z^{k+1} - z^k \rightarrow 0.$$

Substituting into

$$\|x^{k+1} - x^k\| \leq \|u^k - u^{k-1}\| + \|A^\top\| \|y^k - y^{k-1}\| + \frac{1}{t} \|w^k - w^{k-1}\| + \frac{1}{t} \|A^\top\| \|z^k - z^{k-1}\|. \quad (56)$$

we obtain

$$x^{k+1} - x^k \rightarrow 0.$$

Thus all blocks are asymptotically regular:

$$x^{k+1} - x^k \rightarrow 0, \quad u^{k+1} - u^k \rightarrow 0, \quad y^{k+1} - y^k \rightarrow 0, \quad w^{k+1} - w^k \rightarrow 0, \quad z^{k+1} - z^k \rightarrow 0.$$

C.7 Convergence Rates

D Chambolle-Pock

D.1 Convergence analysis for Chambolle-Pock in Section 3.5

Let (x^*, y^*) be a saddle point of the primal-dual formulation (11). Here, x^* represents a mathematical minimizer of our chosen objective function, and y^* is the corresponding optimal dual variable. Formally, (x^*, y^*) satisfies the saddle point condition for the Lagrangian:

$$\langle Ax^*, y \rangle + f(x^*) - g^*(y) \leq \langle Ax^*, y^* \rangle + f(x^*) - g^*(y^*) \leq \langle Ax, y^* \rangle + f(x) - g^*(y^*)$$

for all x, y . From this, the first-order optimality conditions dictate:

$$-A^\top y^* \in \partial f(x^*) \quad \text{and} \quad Ax^* \in \partial g^*(y^*).$$

By definition of the proximal operator (the resolvent of the subdifferential), the updates of the Chambolle-Pock algorithm can be rewritten in terms of subdifferential inclusions:

$$\frac{x^{k-1} - x^k}{t} - A^\top y^k \in \partial f(x^k) \quad \text{and} \quad \frac{y^{k-1} - y^k}{s} + Az^{k-1} \in \partial g^*(y^k).$$

Both f and g^* are closed, proper, convex functions, and hence their subdifferentials are monotone operators. Apply the monotonicity of the subdifferential² to the optimal points at the k -th iterate:

$$\begin{aligned} \left\langle \frac{x^{k-1} - x^k}{t} - A^\top (y^k - y^*), x^k - x^* \right\rangle &\geq 0, \\ \left\langle \frac{y^{k-1} - y^k}{s} + A(z^{k-1} - x^*), y^k - y^* \right\rangle &\geq 0. \end{aligned}$$

Sum these two inequalities and apply standard inner product identities, letting $z^{k-1} := 2x^{k-1} - x^{k-2}$:

$$\left\langle \frac{x^{k-1} - x^k}{t} - A^\top (y^k - y^*), x^k - x^* \right\rangle + \left\langle \frac{y^{k-1} - y^k}{s} + A(z^{k-1} - x^*), y^k - y^* \right\rangle$$

²i.e. $\langle u - v, x - y \rangle \geq 0$ for $u \in \partial f(x)$ and $v \in \partial f(y)$.

$$\begin{aligned}
&= \frac{1}{t} \langle x^{k-1} - x^k, x^k - x^\star \rangle + \frac{1}{s} \langle y^{k-1} - y^k, y^k - y^\star \rangle & (\dagger) \\
&\quad - \langle y^k - y^\star, A(x^k - x^\star) \rangle + \langle A(z^{k-1} - x^\star), y^k - y^\star \rangle & (\ddagger) \\
&\geq 0, & (57)
\end{aligned}$$

where we used $\langle A^\top u, v \rangle = \langle u, Av \rangle$ to group up the matrix products in terms of A . Apply the inner product identity $2\langle a - b, b - c \rangle = \|a - c\|^2 - \|a - b\|^2 - \|b - c\|^2$ to the first two terms ($\dagger$):

$$\begin{aligned}
(\dagger) &= \frac{1}{2t} (\|x^{k-1} - x^\star\|^2 - \|x^{k-1} - x^k\|^2 - \|x^k - x^\star\|^2) \\
&\quad + \frac{1}{2s} (\|y^{k-1} - y^\star\|^2 - \|y^{k-1} - y^k\|^2 - \|y^k - y^\star\|^2). & (58)
\end{aligned}$$

Meanwhile, for next two terms ($\ddagger$):

$$\begin{aligned}
-\langle y^k - y^\star, A(x^k - x^\star) \rangle + \langle y^k - y^\star, A(z^{k-1} - x^\star) \rangle &= \langle y^k - y^\star, A(z^{k-1} - x^\star) - A(x^k - x^\star) \rangle, \\
&= \langle y^k - y^\star, A(z^{k-1} - x^k) \rangle. & (59)
\end{aligned}$$

Recall $z^{k-1} = 2x^{k-1} - x^{k-2} = x^{k-1} + (x^{k-1} - x^{k-2})$ to rewrite the difference:

$$z^{k-1} - x^k = -(x^k - x^{k-1}) + (x^{k-1} - x^{k-2}).$$

Distributing this back into the inner product (59) yields:

$$\begin{aligned}
\langle y^k - y^\star, A(z^{k-1} - x^k) \rangle &= \langle y^k - y^\star, A(-(x^k - x^{k-1}) + (x^{k-1} - x^{k-2})) \rangle \\
&= -\langle y^k - y^\star, A(x^k - x^{k-1}) \rangle + \langle y^k - y^\star, A(x^{k-1} - x^{k-2}) \rangle.
\end{aligned}$$

To create a telescoping sum across iterations, we split the $y^k - y^\star$ component according to $y^k - y^\star = (y^k - y^{k-1}) + (y^{k-1} - y^\star)$:

$$\begin{aligned}
\langle y^k - y^\star, A(x^{k-1} - x^{k-2}) \rangle &= \langle y^k - y^{k-1}, A(x^{k-1} - x^{k-2}) \rangle \\
&\quad + \langle y^{k-1} - y^\star, A(x^{k-1} - x^{k-2}) \rangle.
\end{aligned}$$

Combining these pieces, the original ($\ddagger$) term expands to three inner products:

$$\begin{aligned}
(\ddagger) &= -\langle y^k - y^\star, A(x^k - x^{k-1}) \rangle \\
&\quad + \langle y^{k-1} - y^\star, A(x^{k-1} - x^{k-2}) \rangle \\
&\quad + \langle y^k - y^{k-1}, A(x^{k-1} - x^{k-2}) \rangle & (60)
\end{aligned}$$

Return to our main inequality (57). Substitute the expanded components ($\dagger$) in (58) and ($\ddagger$) in (60) and rearrange the result such that all variables at the current iteration k are on the LHS of (57) and all variables at previous iterations are on the RHS:

$$\begin{aligned}
&\underbrace{\frac{1}{2t} \|x^k - x^\star\|^2 + \frac{1}{2s} \|y^k - y^\star\|^2 + \langle y^k - y^\star, A(x^k - x^{k-1}) \rangle}_{(A)} \\
&\leq \underbrace{\frac{1}{2t} \|x^{k-1} - x^\star\|^2 + \frac{1}{2s} \|y^{k-1} - y^\star\|^2 + \langle y^{k-1} - y^\star, A(x^{k-1} - x^{k-2}) \rangle}_{(B)} \\
&\quad - \underbrace{\left(\frac{1}{2t} \|x^k - x^{k-1}\|^2 + \frac{1}{2s} \|y^k - y^{k-1}\|^2 - \langle y^k - y^{k-1}, A(x^{k-1} - x^{k-2}) \rangle \right)}_{(C)}. & (61)
\end{aligned}$$

Let E^k denote the entire LHS of (61) at iteration k :

$$E^k := \frac{1}{2t} \|x^k - x^\star\|^2 + \frac{1}{2s} \|y^k - y^\star\|^2 + \langle y^k - y^\star, A(x^k - x^{k-1}) \rangle,$$

such that terms (A) and (B) are concisely written as E^k and E^{k-1} , respectively. Let R^k denote the remainder term (C):

$$R^k := \frac{1}{2t} \|x^k - x^{k-1}\|^2 + \frac{1}{2s} \|y^k - y^{k-1}\|^2 - \langle y^k - y^{k-1}, A(x^{k-1} - x^{k-2}) \rangle.$$

Our massive inequality (61) is written more compactly:

$$E^k \leq E^{k-1} - R^k. \quad (62)$$

Sum the inequality over steps $k = 1, \dots, K$:

$$\sum_{k=1}^K E^k \leq \sum_{k=1}^K E^{k-1} - \sum_{k=1}^K R^k.$$

The form of E^k implies that the first two sums are telescoping, and the previous expression reduces:

$$E^K \leq E^0 - \sum_{k=1}^K R^k. \quad (63)$$

Apply Cauchy-Schwarz and Young's inequality³ to bound the cross term in the definition of R^k as follows:

$$\begin{aligned} \langle y^k - y^{k-1}, A(x^{k-1} - x^{k-2}) \rangle &\leq \|y^k - y^{k-1}\| \cdot \|A\|_2 \cdot \|x^{k-1} - x^{k-2}\|, \\ &= \left(\frac{1}{\sqrt{s}} \|y^k - y^{k-1}\| \right) \left(\sqrt{st} \|A\|_2 \right) \left(\frac{1}{\sqrt{t}} \|x^{k-1} - x^{k-2}\| \right), \\ &\leq \frac{\sqrt{st} \|A\|_2}{2s} \|y^k - y^{k-1}\|^2 + \frac{\sqrt{st} \|A\|_2}{2t} \|x^{k-1} - x^{k-2}\|^2, \\ &= \frac{c}{2s} \|y^k - y^{k-1}\|^2 + \frac{c}{2t} \|x^{k-1} - x^{k-2}\|^2, \end{aligned}$$

where we have set $c := \sqrt{st} \|A\|_2$. The remainder term R^k is therefore given the lower bound:

$$R^k \geq \frac{1}{2s} (1 - c) \|y^k - y^{k-1}\|^2 + \frac{1}{2t} \|x^k - x^{k-1}\|^2 - \frac{c}{2t} \|x^{k-1} - x^{k-2}\|^2.$$

Sum this lower bound from $k = 1, \dots, K$:

$$\sum_{k=1}^K R^k \geq \frac{1-c}{2s} \sum_{k=1}^K \|y^k - y^{k-1}\|^2 + \frac{1}{2t} \sum_{k=1}^K \|x^k - x^{k-1}\|^2 - \frac{c}{2t} \sum_{k=1}^K \|x^{k-1} - x^{k-2}\|^2. \quad (64)$$

To combine the x terms, we shift the index on the final sum, rewriting it as:⁴

$$\sum_{k=1}^K \|x^{k-1} - x^{k-2}\|^2 = 0 + \sum_{k=2}^K \|x^{k-1} - x^{k-2}\|^2 = \sum_{k=1}^{K-1} \|x^k - x^{k-1}\|^2.$$

Substitute this back into the lower bound (64) and group the matching terms from $k = 1, \dots, K-1$:

$$\begin{aligned} \sum_{k=1}^K R^k &\geq \frac{1-c}{2s} \sum_{k=1}^K \|y^k - y^{k-1}\|^2 \\ &\quad + \frac{1-c}{2t} \left(\sum_{k=1}^{K-1} \|x^k - x^{k-1}\|^2 \right) + \frac{1}{2t} \|x^K - x^{K-1}\|^2. \end{aligned} \quad (65)$$

Since each norm in (65) is non-negative, it follows that the entire expression is non-negative whenever $1 - c > 0$, hence the accumulated remainder is itself non-negative $\sum_{k=1}^K R^k \geq 0$ provided $1 - c > 0$. We defined the scalar $c = \sqrt{st} \|A\|_2$, hence $1 - c > 0$ is equivalent to the strict step-size condition:

$$st \|A\|_2^2 < 1.$$

Returning to (63), we have shown that $E^K \leq E^0$. Since K is an arbitrary iteration index, it follows that $E^k \leq E^0$ for any index in the sequence: $0 \leq E^k \leq E^0$ for all $k \geq 1$. Since E^k is

³ $ab \leq \frac{a^2}{2} + \frac{b^2}{2}$.

⁴ If we assume the standard initialization $x^{-1} = x^0$, then term at $k = 1$ vanishes ($\|x^0 - x^{-1}\|^2 = 0$) and all is well in the world.

globally bounded, it follows that the corresponding distances of its arguments relative to the optimizer $\|x^k - x^*\|$ and $\|y^k - y^*\|$ is found within a bounded region of Euclidean space. Thus the sequence of iterates (x^k, y^k) is bounded.

To complete the proof of convergence, we analyze the asymptotic behavior of the (bounded) sequence. Rearrange (63) to see $\sum_{k=1}^K R^k \leq E^0 - E^K \leq E^0$, and take the limit as $K \rightarrow \infty$ to find:

$$\sum_{k=1}^{\infty} R^k \leq E^0 < \infty.$$

Substitute the shifted lower bound (65) into this infinite sum. Whenever the strict step-size condition is met, $\|A\|_2^2 < 1$, the infinite series of R^k contains strictly non-negative terms. Since this series converges to a finite value (i.e. some non-negative value $\leq E^0$), it follows that the individual terms R^k must vanish in the limit, and in particular its constituent terms must also vanish:

$$\lim_{k \rightarrow \infty} \|x^k - x^{k-1}\|^2 = 0, \quad \text{and} \quad \lim_{k \rightarrow \infty} \|y^k - y^{k-1}\|^2 = 0. \quad (66)$$

We have already proven that the sequence of iterates (x^k, y^k) is globally bounded, thus Bolzano-Weierstrass applies and guarantees the existence of at least one convergent subsequence. Let (x^{k_m}, y^{k_m}) denote such a subsequence and let $(\bar{x}, \bar{y})$ denote its limiting point. It follows from (66) that the preceding iterates (x^{k_m-1}, y^{k_m-1}) must also converge to the same limit point $(\bar{x}, \bar{y})$, meaning that $(\bar{x}, \bar{y})$ is a fixed point of the algorithm's iterates. The subdifferential optimality conditions imply that its roots coincide precisely with such fixed points, and hence $(\bar{x}, \bar{y})$ is in fact a valid saddle point of the original min-max problem: $(\bar{x}, \bar{y}) = (x^*, y^*)$.

E Algorithm Pseudocode

Algorithm 1 presents a pseudocode sketch of the PDR algorithm, Algorithm 2 presents the PDDR algorithm, Algorithm 3 presents the ADMM algorithm, and Algorithm 4 the Chambolle-Pock algorithm.

Algorithm 1: Primal Douglas-Rachford Splitting

Input: Blurred image b , step size t , relaxation parameter ρ

Output: Deblurred image x_{sol}

```

1 Initialize:  $z_1^0 \in \mathbb{R}^n, z_2^0 \in \mathbb{R}^{3n}$ ;
2 for  $k = 1, 2, \dots$  do
3    $x^k = \text{prox}_{tf}(z_1^{k-1})$ ;
4    $y^k = \text{prox}_{tg}(z_2^{k-1})$ ;
5    $u^k = (I + A^\top A)^{-1}(2x^k - z_1^{k-1} + A^\top(2y^k - z_2^{k-1}))$ ;
6    $v^k = Au^k$ ;
7    $z_1^k = z_1^{k-1} + \rho(u^k - x^k)$ ;
8    $z_2^k = z_2^{k-1} + \rho(v^k - y^k)$ ;
9 end
10 return  $x_{\text{sol}} = \text{prox}_{tf}(z_1^k)$ ;

```

F Software Package Documentation

Entry-point: deblur

```

x = deblur(problem, algorithm, xinitial, kernel, b)
x = deblur(problem, algorithm, xinitial, kernel, b, i)

```

Inputs

Algorithm 2: Primal-Dual Douglas-Rachford Splitting

Input: Blurred image b , step size t , relaxation parameter ρ **Output:** Deblurred image x_{sol}

```
1 Initialize:  $p^0 \in \mathbb{R}^n, q^0 \in \mathbb{R}^{3n}$ ;  
2 for  $k = 1, 2, \dots$  do  
3    $x^k = \text{prox}_{tf}(p^{k-1})$ ;  
4    $z^k = \text{prox}_{tg^*}(q^{k-1})$ ;  
5    $w^k = (I + t^2 A^\top A)^{-1} (2x^k - p^{k-1} + tA^\top (2z^k - q^{k-1}))$ ;  
6    $v^k = (2z^k - q^{k-1}) - tAw^k$ ;  
7    $p^k = p^{k-1} + \rho(w^k - x^k)$ ;  
8    $q^k = q^{k-1} + \rho(v^k - z^k)$ ;  
9 end  
10 return  $x_{\text{sol}} = \text{prox}_{tf}(p^k)$ ;
```

Algorithm 3: Alternating Direction Method of Multipliers (ADMM)

Input: Blurred image b , step size t , relaxation parameter ρ **Output:** Deblurred image x_{sol}

```
1 Initialize:  $u^0, y^0, w^0, z^0$ ;  
2 for  $k = 1, 2, \dots$  do  
3    $x^k = (I + A^\top A)^{-1} (u^{k-1} + A^\top y^{k-1} - \frac{1}{t}(w^{k-1} + A^\top z^{k-1}))$ ;  
4    $u^k = \text{prox}_{t^{-1}f}(\rho x^k + (1 - \rho)u^{k-1} + \frac{w^{k-1}}{t})$ ;  
5    $y^k = \text{prox}_{t^{-1}g}(\rho Ax^k + (1 - \rho)y^{k-1} + \frac{z^{k-1}}{t})$ ;  
6    $w^k = w^{k-1} + t(x^k - u^k)$ ;  
7    $z^k = z^{k-1} + t(Ax^k - y^k)$ ;  
8 end  
9 return  $x_{\text{sol}} = (I + A^\top A)^{-1} (u^k + A^\top y^k - \frac{1}{t}(w^k + A^\top z^k))$ ;
```

Algorithm 4: Chambolle-Pock algorithm

Input: Blurred image b , primal step size t , dual step size s **Output:** Deblurred image x_{sol}

```
1 Initialize:  $x^0, y^0, z^0$ ;  
2 for  $k = 1, 2, \dots$  do  
3    $y^k = \text{prox}_{sg^*}(y^{k-1} + sAz^{k-1})$ ;  
4    $x^k = \text{prox}_{tf}(x^{k-1} - tA^\top y^k)$ ;  
5    $z^k = 2x^k - x^{k-1}$ ;  
6 end  
7 return  $x_{\text{sol}} = x^k$ ;
```

- `problem` — 'l1', 'l2', or 'huber'
- `algorithm` — 'PDR', 'PDDR', 'ADMM', or 'CP' (case-insensitive)
- `xinitial` — $(H \times W)$ initial guess; pass `[]` to start from b
- `kernel` — $(k_H \times k_W)$ point-spread function
- `b` — $(H \times W)$ blurred and noisy observation, values in $[0, 1]$
- `i` — optional parameter struct; any subset of fields in Table 1 may be specified

Output — $(H \times W)$ double, reconstructed image in $[0, 1]$.

Installation

Run from the `project/app/` directory, or add the following to the MATLAB path:

```
addpath('project/app')
addpath('project/app/algorithms')
addpath('project/app/prox_operators')
```

Example

```
psf = fspecial('gaussian', 9, 2);
b = imfilter(im2double(imread('cameraman.tif')), psf,
'circular');
b = b + 0.02 * randn(size(b));

x = deblur('l2', 'PDDR', [], psf, b);           % defaults

i.gamma = 0.008; i.maxiter = 300;
x = deblur('l2', 'ADMM', [], psf, b, i);       % custom
params
```

Graphical Interface

A MATLAB UIFigure application (`DeblurApp.m`) wraps the same backend. Load an image, configure blur and noise, select algorithm and fidelity type, then press **Run**. The app displays the reconstructed image, convergence curve, and PSNR/SSIM scores side-by-side. Launch with `DeblurApp` from the MATLAB command window.

F.1 Convergence behavior of the default settings

With the shipped defaults ($\gamma_{\ell_1} = 0.006$, $\gamma_{\ell_2} = 0.012$, `maxiter` = 500, `tol` = 10^{-4}), the two fidelity models behave asymmetrically. Running `run_defaults.m` on the corrupted cameraman image (Gaussian blur 9×9 , $\sigma = 2.0$, Gaussian noise level 0.02) we observed:

- **ℓ_2 problem.** Primal Douglas-Rachford, Primal-Dual Douglas-Rachford, and Chambolle-Pock all met the relative-change tolerance well before the iteration cap, terminating at iterations 172, 155, and 185 respectively. ADMM grazed the cap with relative change within a small multiple of `tol`.
- **ℓ_1 problem.** All four algorithms exhausted the full 500 iterations with relative change plateauing near 10^{-4} , which is close to, but not crossing the tolerance.

This is consistent with the non-smoothness of the ℓ_1 data term, which removes the local strong-convexity that accelerates the ℓ_2 iterates in a neighborhood of the optimum; only the $O(1/k)$ rate guaranteed by splitting theory remains. The practical implication is that the ℓ_1 model trades faster per-iteration robustness to impulsive noise for a markedly slower tail. Callers who want a definitive "converged" flag on the ℓ_1 problem have three options:

1. Raise the iteration budget, e.g. `i.maxiter` = 2000, which typically suffices for all four algorithms on corruption levels in the range we tested.

2. Loosen the stopping tolerance to `i.tol = 5e-4`; the resulting reconstructions are visually indistinguishable from the `tol = 10-4` counterparts in our tests.
3. Switch to the Huber fidelity (`problem = 'huber'`, with optional `i.delta` controlling the quadratic-to-linear transition default = 0.1) as a smoothed surrogate that preserves impulse-noise robustness while restoring local smoothness and therefore faster asymptotic convergence.

The iteration cap is *not* a failure mode: the intermediate iterates at $k = 500$ are already near-optimal in objective value and visually converged; only the relative-change stopping criterion has not formally certified termination.

G Experiments: Supplementary Material and Results

G.1 ℓ_2 and Motion Blur Reconstructions for Section 6.1

Figure 12 presents the image reconstructions for all four algorithms under the ℓ_2 model with Gaussian blur (15×15 and $\sigma = 2$) and Gaussian noise ($\sigma = 0.01$).

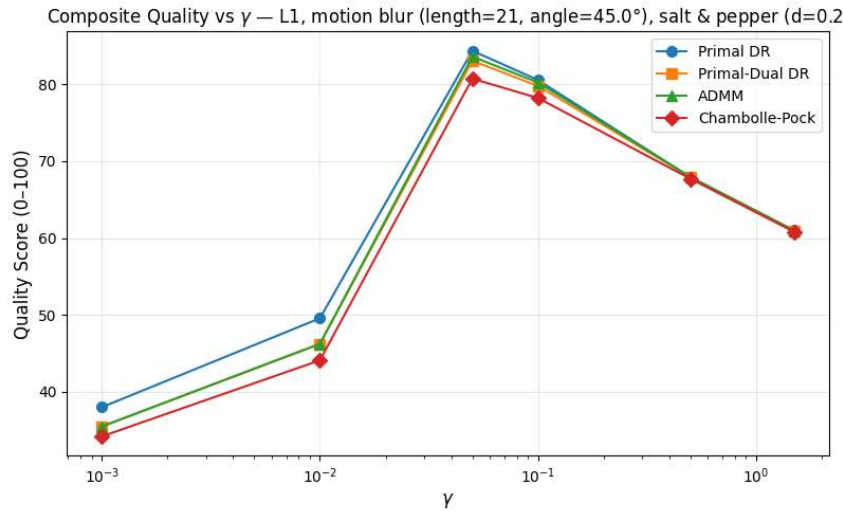


Figure 8: Composite quality score as a function of γ for the four reconstruction algorithms under the ℓ_1 formulation with motion blur of length 21 and angle 45° , with salt-and-pepper noise at level $d = 0.2$. All methods achieve their best performance for intermediate values of γ , with performance deteriorating for very small or very large regularization.



Figure 9: Reconstructed images for the four algorithms across different values of γ under the ℓ_1 formulation with motion blur of length 21 and angle 45° , and salt-and-pepper noise at level $d = 0.2$. Small values of γ leave substantial noise in the reconstruction, intermediate values provide the best visual quality, and large values oversmooth the image and introduce blur.

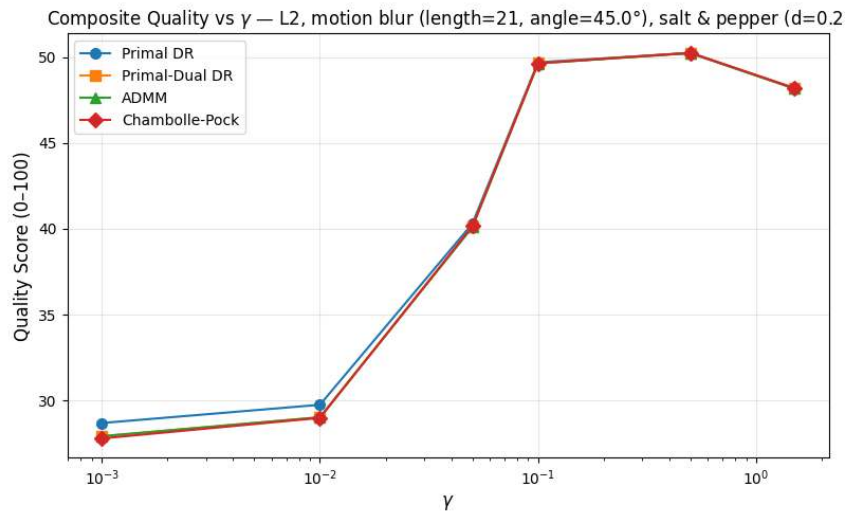


Figure 10: Composite quality score as a function of γ for the four reconstruction algorithms under the L_2 formulation with motion blur of length 21 and angle 45° , with salt-and-pepper noise at level $d = 0.2$. Reconstruction quality improves as γ increases from very small values, reaches its best performance for intermediate to large values of γ , and then slightly declines for the largest value considered.



Figure 11: Reconstructed images for the four algorithms across different values of γ under the L_2 formulation with motion blur of length 21 and angle 45° , and salt-and-pepper noise at level $d = 0.2$. For this setting, small values of γ leave substantial noise in the reconstruction, while larger values improve denoising but produce oversmoothed and blurry reconstructions.



Figure 12: ℓ_2 reconstruction. The actual figure has been added in the list of files accessible through the file and folder menu to the right.

G.2 Golden section search: supplementary figures

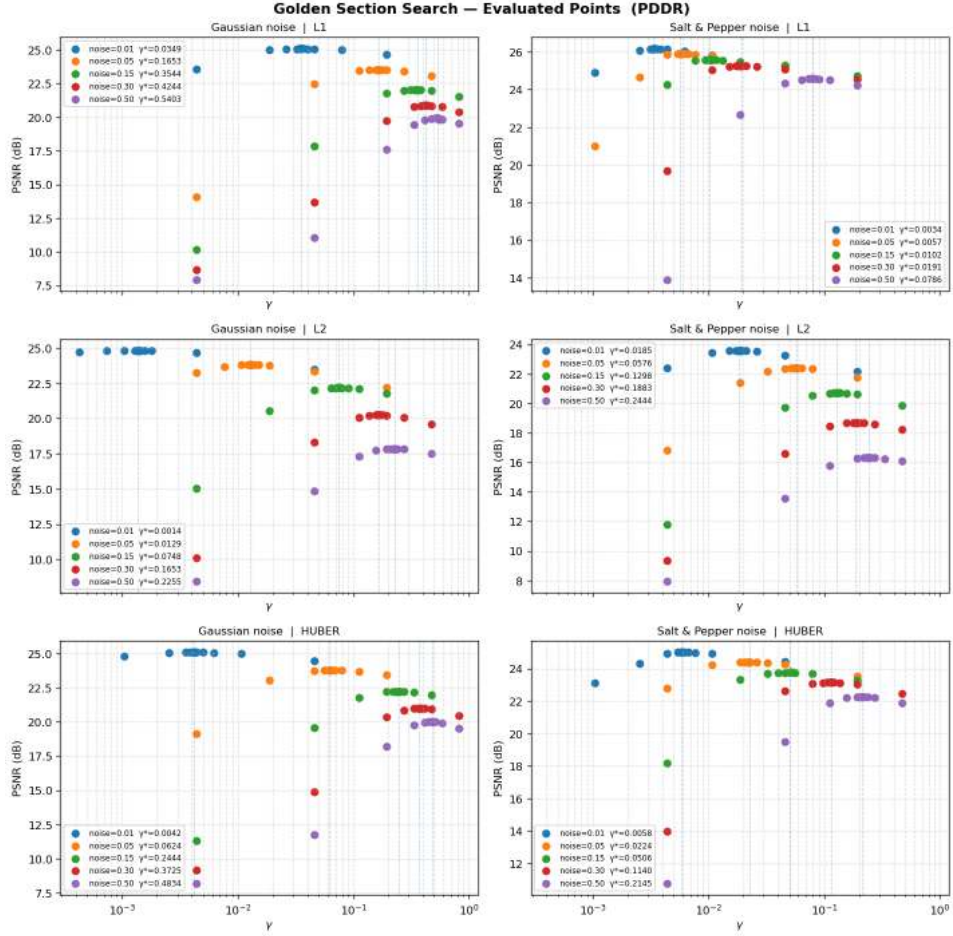


Figure 13: Evaluated (γ, PSNR) points during golden section search in PDDR for each noise level and problem variant. Dashed vertical lines mark γ^* . The sparse, non-uniform spacing illustrates the bracketing structure of the search.

When the noise type is known, the oracle model (ℓ_1 for salt-and-pepper, ℓ_2 for Gaussian) consistently matches or outperforms the Huber model. For instance, ℓ_1 achieves 24.59 dB versus Huber’s 22.27 dB at $\sigma = 0.50$ under salt-and-pepper noise. Huber’s γ^* and PSNR always sit between those of ℓ_1 and ℓ_2 , confirming its role as a robust fallback for unknown or mixed noise rather than a strict improvement over either pure model.

G.3 Gamma vs. image resolution: supplementary figures

The following Figures 15, 16, 17, 8, and 9 support the discussion in Section 6.3.

G.4 Chambolle-Pock step-size experiment in Section 6.5

Figure 18 shows the sample cameraman and manWithHat images before corruption is applied and after Gaussian/motion blurring and salt-and-pepper/Gaussian noise.

Gaussian blur. Figures 19 and 20 illustrates the quality score evaluations and the image reconstructions (respectively) for the cameraman image under Gaussian blurring, while Figures 21 and 22 show the scores and reconstructions for the manWithHat test image, exhibiting qualitatively similar results.

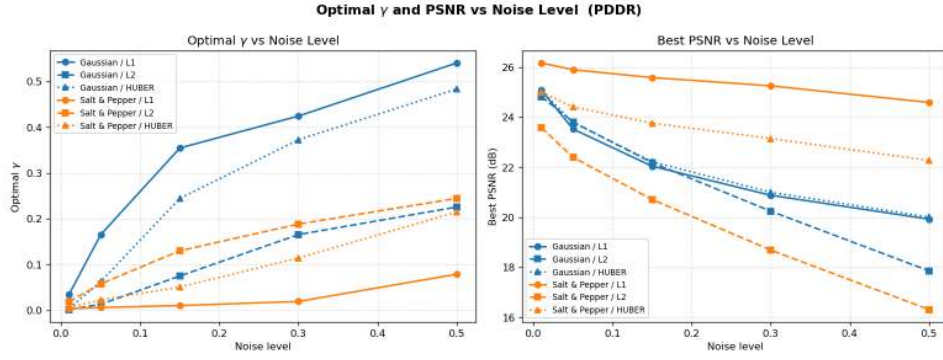


Figure 14: Left: optimal γ^* as a function of noise level for all noise-type/model combinations. Right: corresponding best PSNR (dB). Both are monotone in noise level.

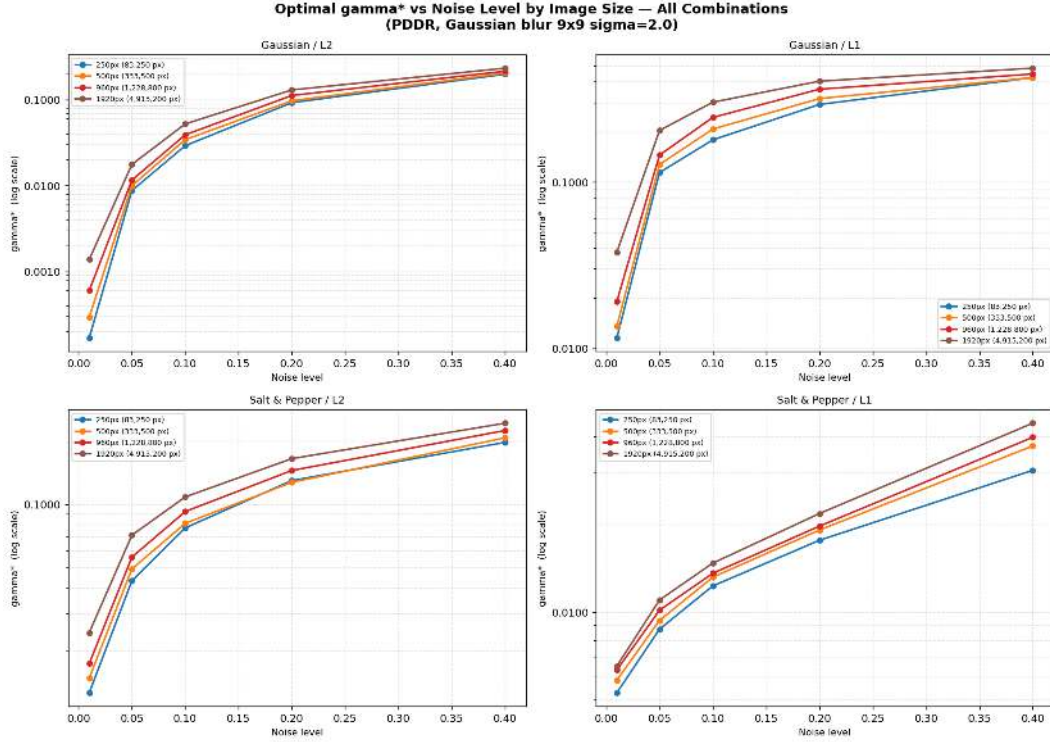


Figure 15: Optimal γ^* vs. noise level for each image resolution, across all four noise-type/model combinations. γ^* increases with both noise level and resolution; the effect is strongest for Gaussian noise under the ℓ_2 model.

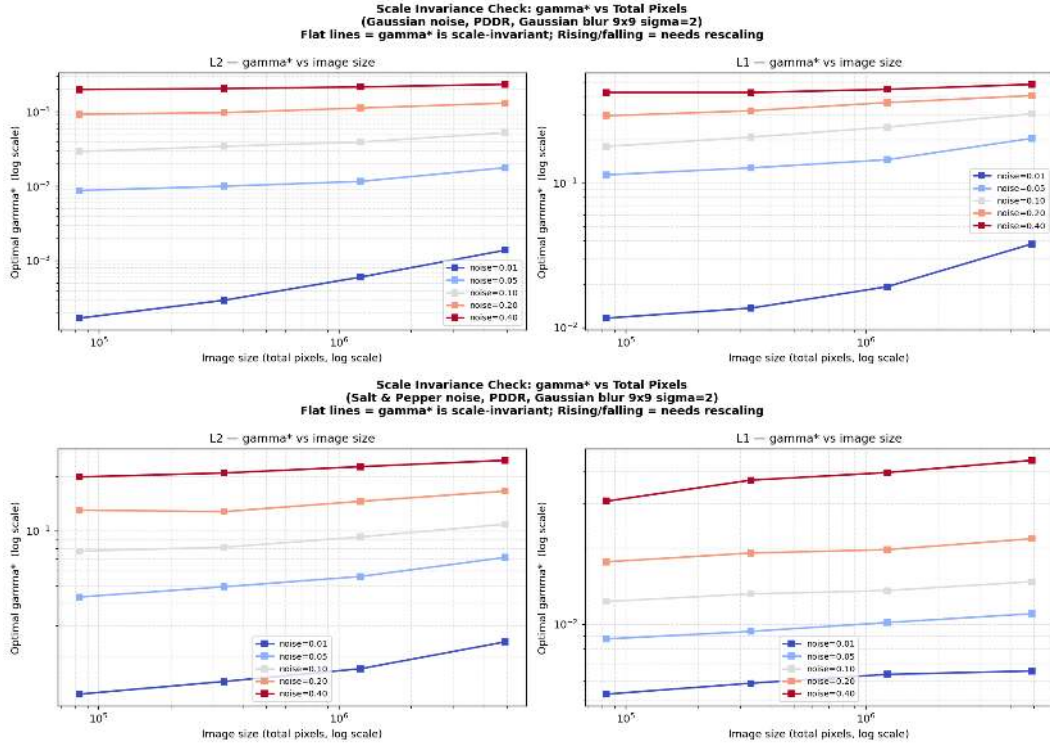


Figure 16: Log-log plot of γ^* vs. total pixel count for each noise level. Flat lines would indicate scale invariance. Left (Gaussian): lines rise with slope ≈ 0.51 , confirming $\gamma^* \propto \sqrt{N}$. Right (S&P): lines under the ℓ_1 model are nearly flat; the ℓ_2 model shows moderate growth.

Motion blur. Figures 23 and 24 illustrates the quality score evaluations and the image reconstructions (respectively) for the cameraman image under motion blurring, while Figures 25 and 26 show the scores and reconstructions for the manWithHat test image, exhibiting qualitatively similar results to the cameraman experiments.

G.5 Parameter tuning experiment for Section 6.6

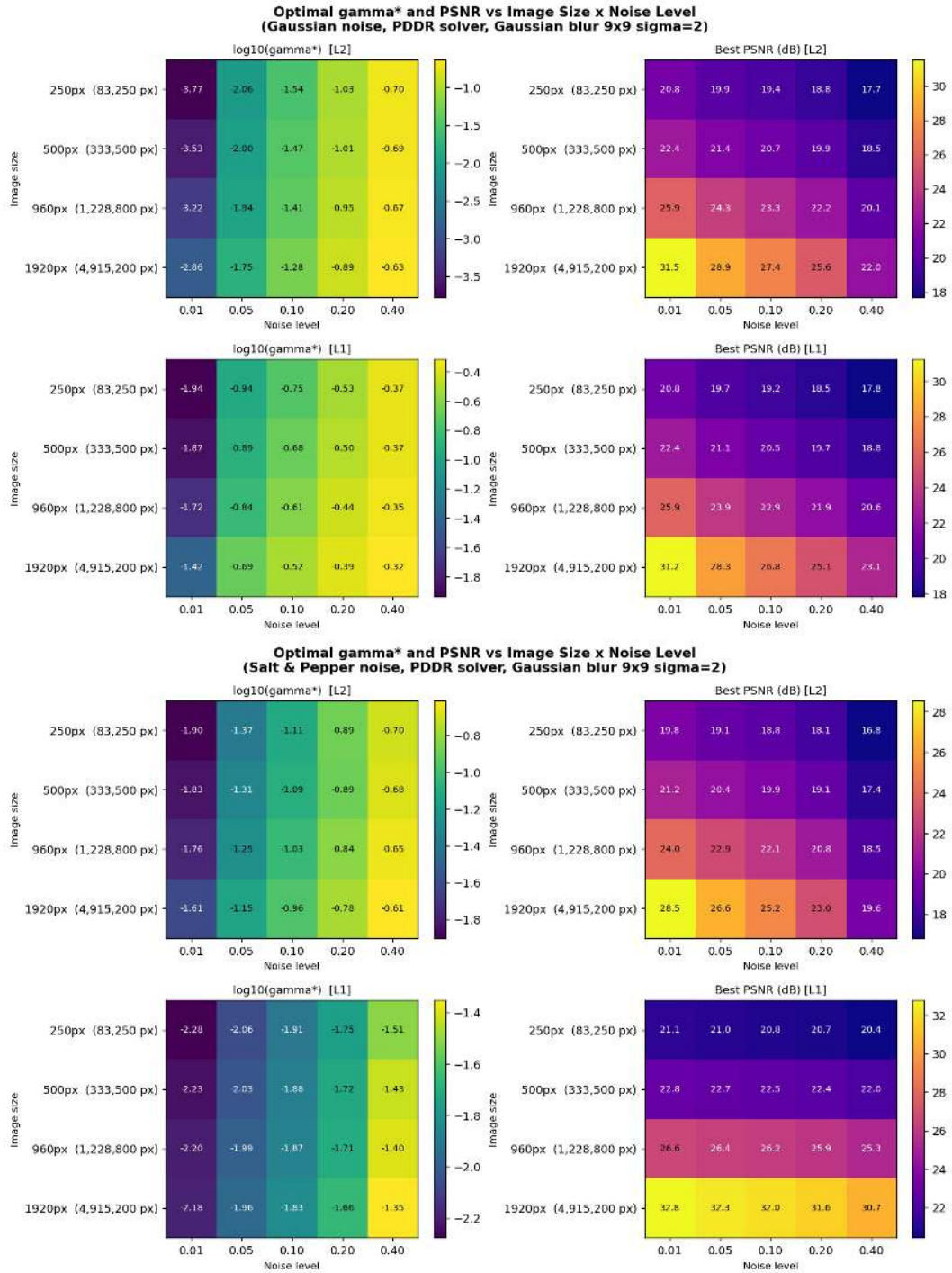


Figure 17: Heatmaps of $\log_{10}(\gamma^*)$ (left column of each panel) and best PSNR in dB (right column) across image size $\times$ noise level, for Gaussian (left) and salt-and-pepper (right) noise.



Figure 18: Original images and their corruptions passed to the Chambolle-Pock step-size experiment.

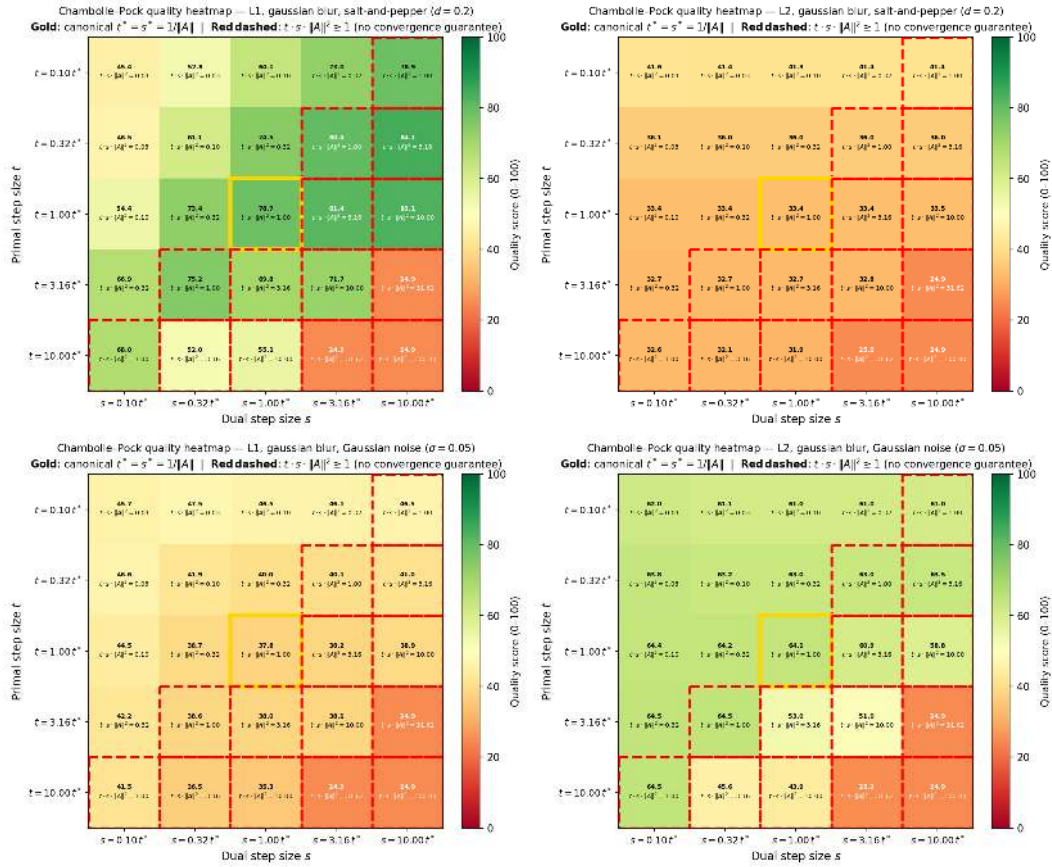


Figure 19: Gaussian blurring (cameraman): Reconstruction quality under the ℓ_1 model (left column) and ℓ_2 model (right column), with salt-and-pepper noise (top row) and Gaussian noise (bottom row).

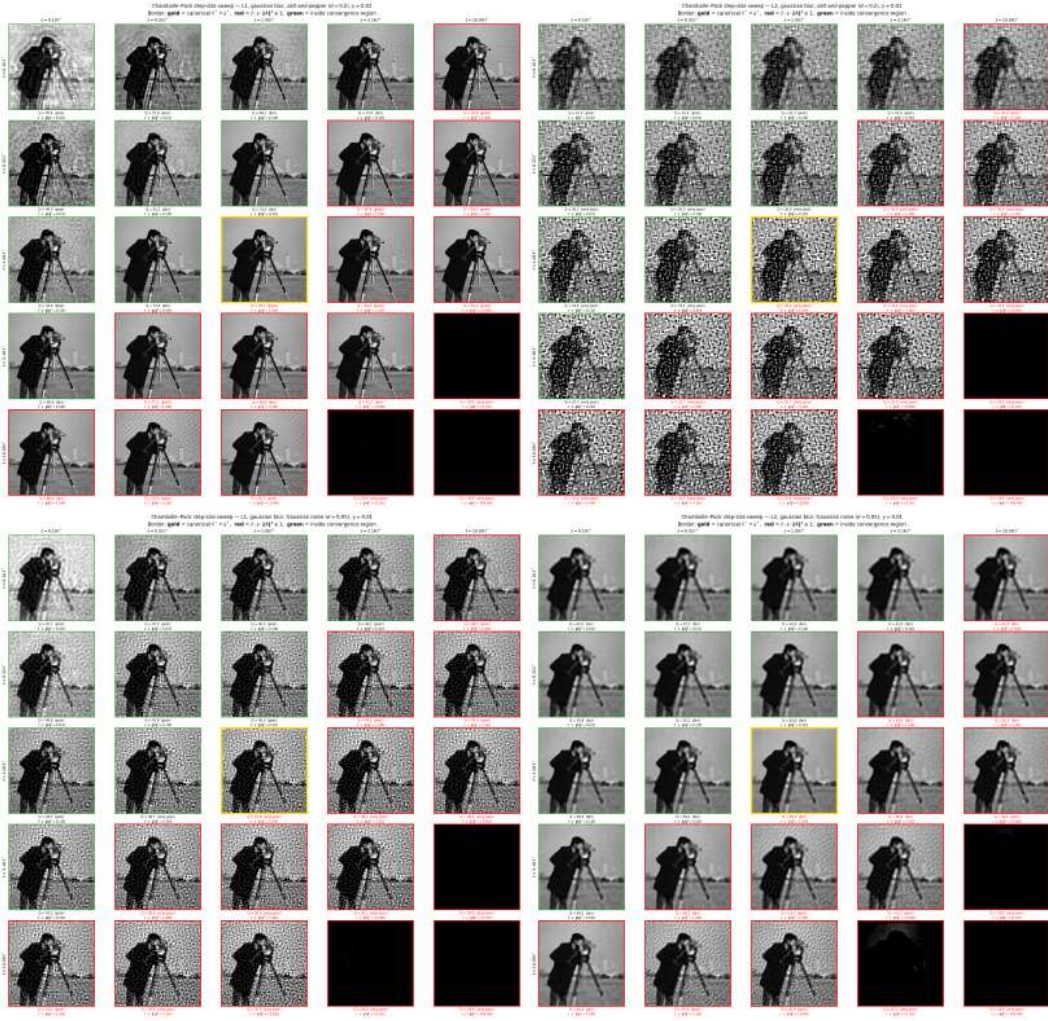


Figure 20: Gaussian blurring (cameraman): Image reconstructions under the ℓ_1 model (left column) and ℓ_2 model (right column), with salt-and-pepper noise (top row) and Gaussian noise (bottom row).

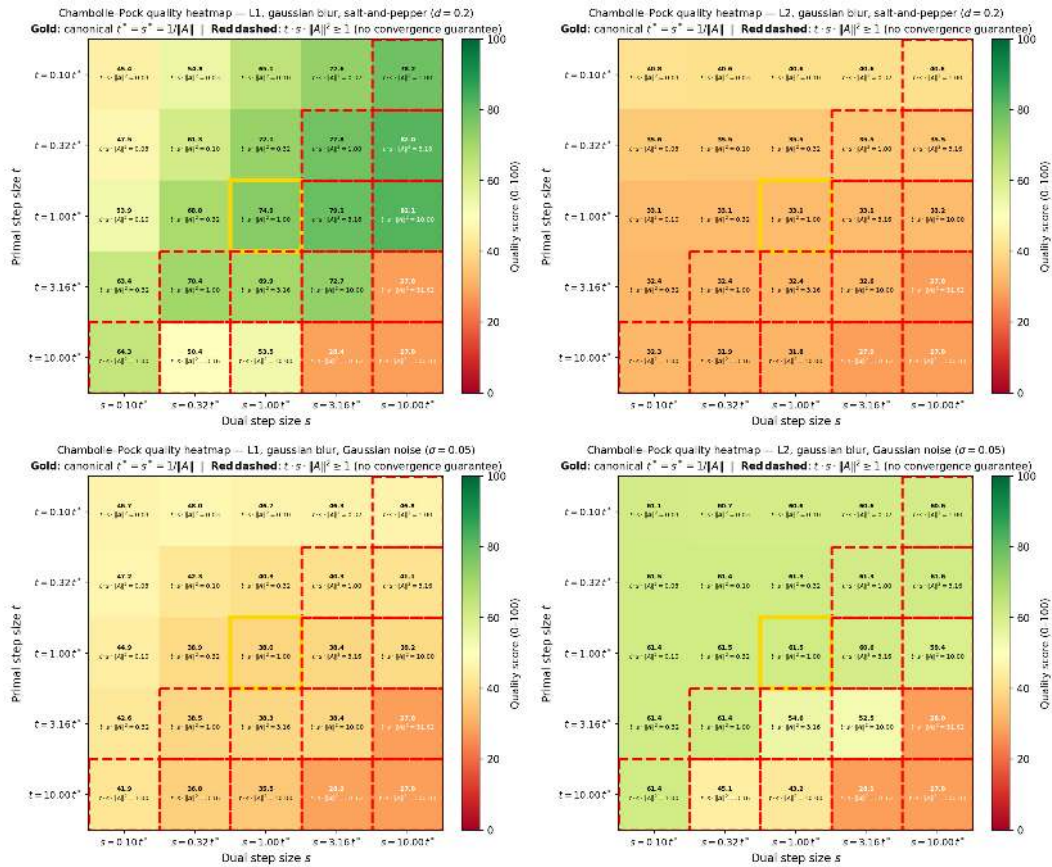


Figure 21: Gaussian blurring (manWithHat): Reconstruction quality under the ℓ_1 model and ℓ_2 model (right column), with salt-and-pepper noise (top row) and Gaussian noise (bottom row).



Figure 22: Gaussian blurring (manWithHat): Image reconstructions under the ℓ_1 model (left column) and ℓ_2 model (right column), with salt-and-pepper noise (top row) and Gaussian noise (bottom row).

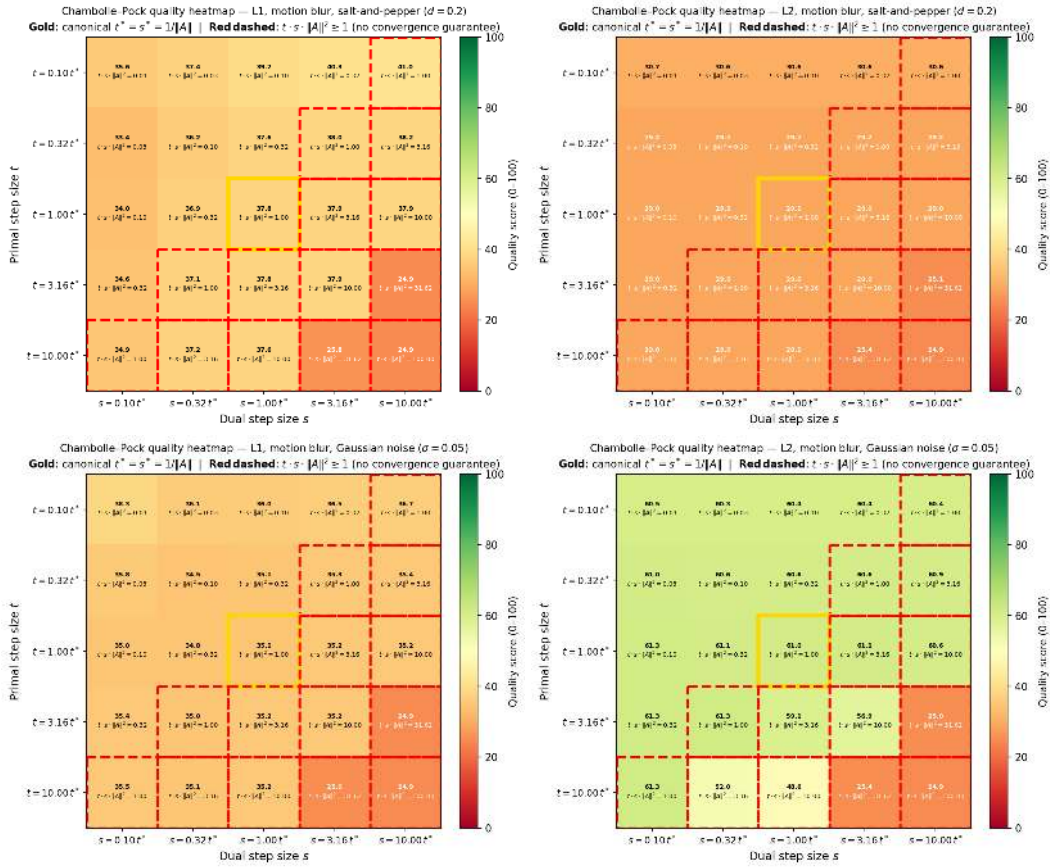


Figure 23: Motion blurring (cameraman): Reconstruction quality under the ℓ_1 model (left column) and ℓ_2 model (right column), with salt-and-pepper noise (top row) and Gaussian noise (bottom row).

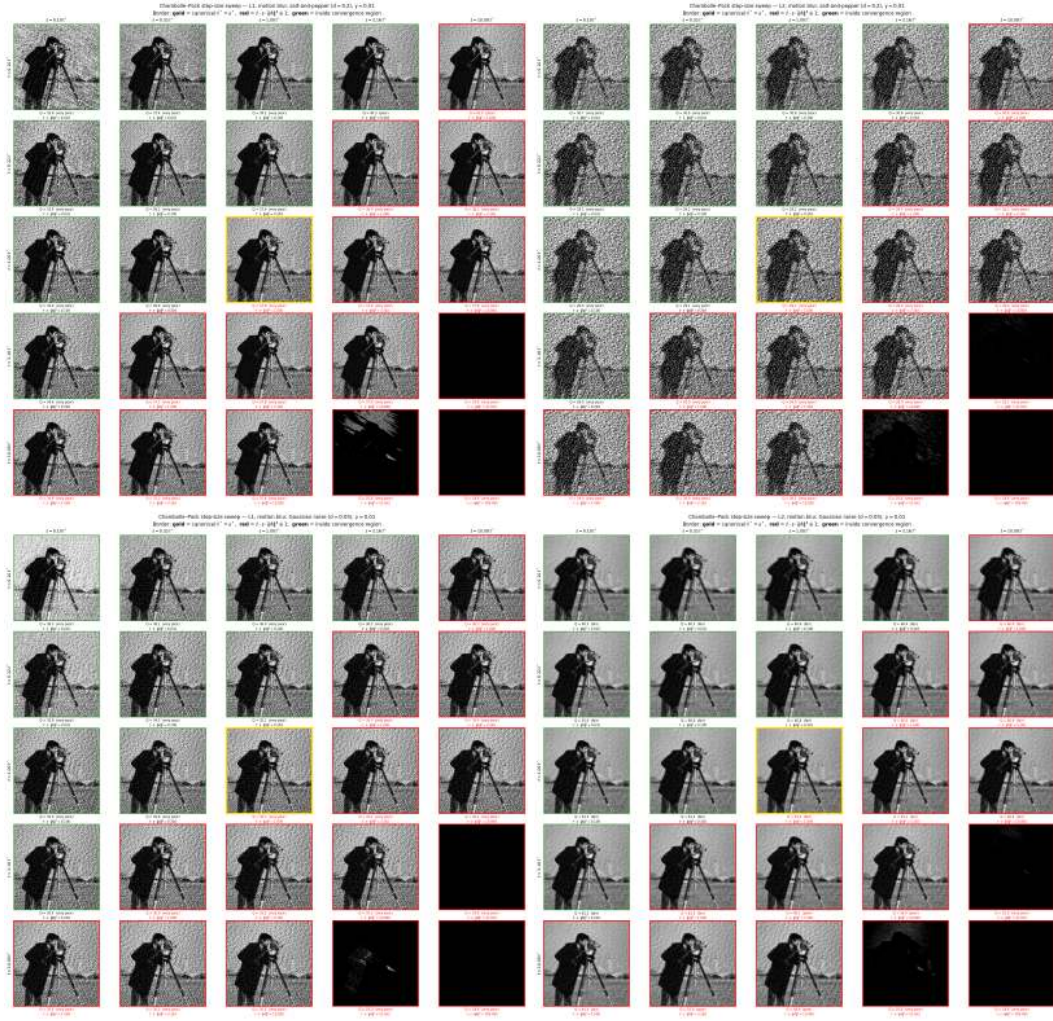


Figure 24: Motion blurring (cameraman): Image reconstructions under the ℓ_1 model (left column) and ℓ_2 model (right column), with salt-and-pepper noise (top row) and Gaussian noise (bottom row).

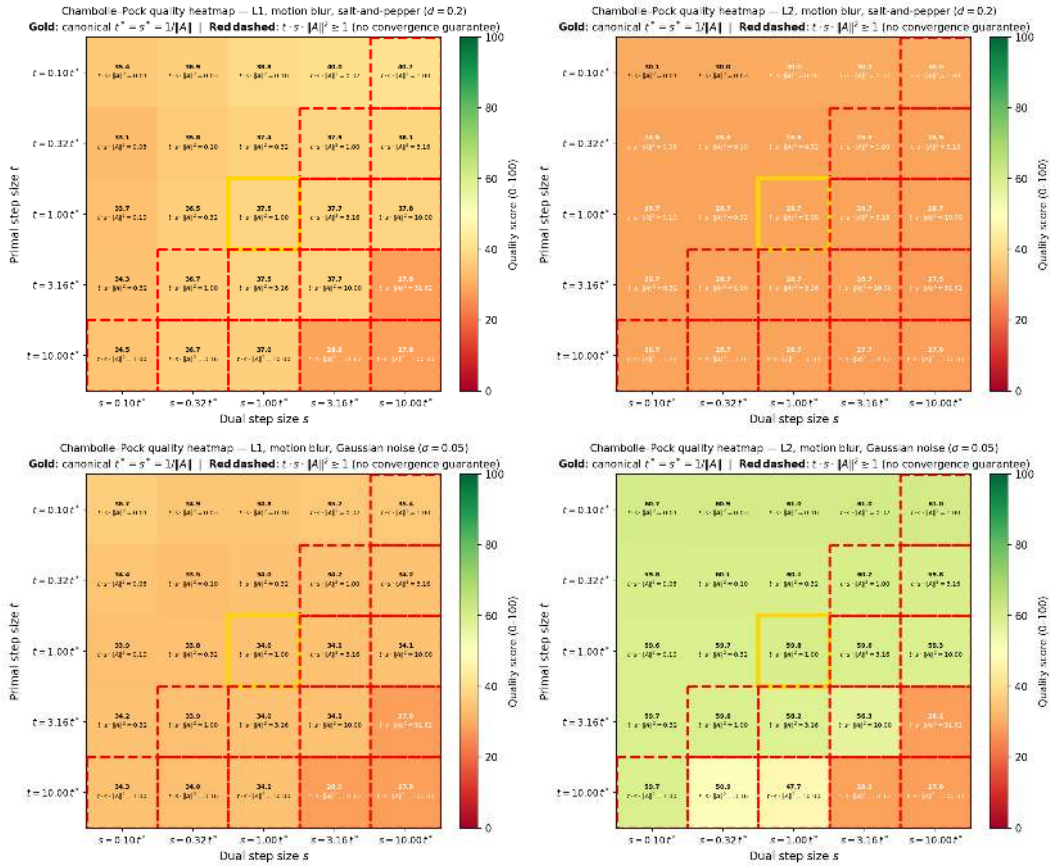


Figure 25: Motion blurring (manWithHat): Reconstruction quality under the ℓ_1 model (left column) and ℓ_2 model (right column), with salt-and-pepper noise (top row) and Gaussian noise (bottom row).



Figure 26: Motion blurring (manWithHat): Image reconstructions under the ℓ_1 model (left column) and ℓ_2 model (right column), with salt-and-pepper noise (top row) and Gaussian noise (bottom row).

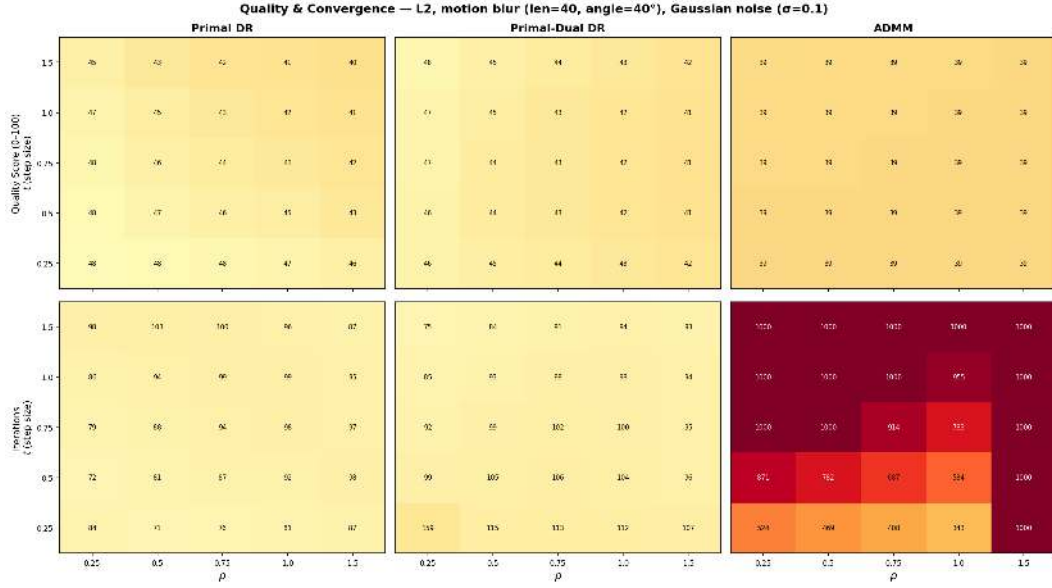


Figure 27: Quality score and iterations to convergence as a function of step size t and relaxation parameter ρ for each algorithm, under motion blur (len=40, angle=40°) with Gaussian noise ($\sigma = 0.1$) and ℓ_2 formulation.

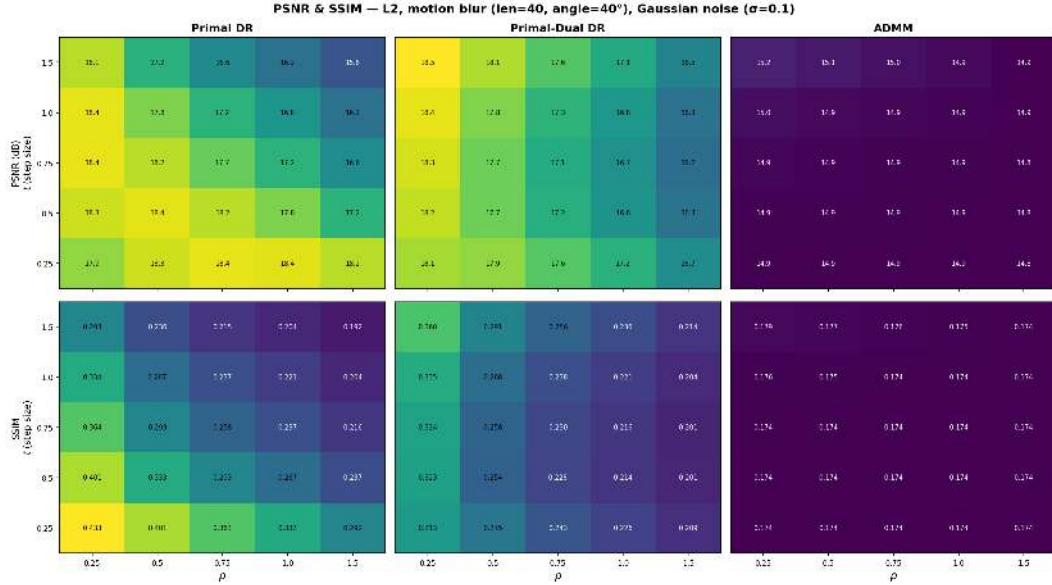


Figure 28: PSNR and SSIM as a function of step size t and relaxation parameter ρ for each algorithm, under the same experimental setup as Figure 27.